



OECD Regulatory Policy Working Papers No. 26

The spectrum of regulatory
models and its application
to the digital economy

**Becky King,
Natalie Cohen,
Anna Pietikäinen,
Wiktor Samek**

<https://dx.doi.org/10.1787/b7456763-en>

The spectrum of regulatory models and its application to the digital economy

This work is issued under the responsibility of the Secretary-General of the OECD and does not necessarily reflect the official views of OECD Member countries.

This document, as well as any data and map included herein, are without prejudice to the status of or sovereignty over any territory, to the delimitation of international frontiers and boundaries and to the name of any territory, city or area.

OECD Regulatory Policy Working Papers are published on https://www.oecd.org/en/publications/oecd-regulatory-policy-working-papers_24140996.html

© OECD 2026



Attribution 4.0 International (CC BY 4.0)

This work is made available under the Creative Commons Attribution 4.0 International licence. By using this work, you accept to be bound by the terms of this licence (<https://creativecommons.org/licenses/by/4.0/>).

Attribution – you must cite the work.

Translations – you must cite the original work, identify changes to the original and add the following text: *In the event of any discrepancy between the original work and the translation, only the text of original work should be considered valid.*

Adaptations – you must cite the original work and add the following text: *This is an adaptation of an original work by the OECD. The opinions expressed and arguments employed in this adaptation should not be reported as representing the official views of the OECD or of its Member countries.*

Third-party material – the licence does not apply to third-party material in the work. If using such material, you are responsible for obtaining permission from the third party and for any claims of infringement.

You must not use the OECD logo, visual identity or cover image without express permission or suggest the OECD endorses your use of the work.

Any dispute arising under this licence shall be settled by arbitration in accordance with the Permanent Court of Arbitration (PCA) Arbitration Rules 2012. The seat of arbitration shall be Paris (France). The number of arbitrators shall be one.

Abstract

Policymakers regulating the digital economy face a difficult question: not only whether to intervene, but how to do so proportionately, balancing innovation against fast-evolving risks. This paper helps navigate that choice. It develops a methodology positioning regulatory models on a spectrum – from command-and-control, through performance-based regulation, market-based mechanisms and co-regulation, to different forms of self-regulation – categorised by their flexibility and the roles public and private actors play across the regulatory process. This enables structured comparison, clarifies trade-offs, and supports design of blended approaches combining different models. The paper offers criteria for model selection based on impact on innovation, market and private actor characteristics, and public institutional readiness. Applied to the digital economy, it finds that governments increasingly rely on blended models, with public actors retaining responsibility for setting regulatory objectives, while delegating the means of achieving them and regulatory delivery to private actors, under public oversight and accountability.

Acknowledgements

This paper was prepared by the OECD Public Governance Directorate (GOV), under the leadership of Elsa Pilichowski, Director, Gillian Dörner, Deputy Director and the direction of Anna Pietikäinen, Head of the Regulatory Policy Division (REG).

The project was led by Wiktor Samek, Natalie Cohen and Becky King, including the development of the methodology for the paper, case collection and analysis, and the drafting of the report. Review and strategic steering were provided by Natalie Cohen, under the general supervision of Anna Pietikäinen. The authors thank, in particular, James Drummond, who initiated the work on the project and contributed greatly to the conceptual framework presented in the paper. The authors are also grateful for the support of colleagues in the Regulatory Governance for Global Challenges Unit, and REG, who provided regular comments that shaped the final version of the report. This includes, in particular, Marianna Karttunen, Christiane Arndt-Basle, Shemsije Jashari, Gábor Kisvárdai, Alberto Morales, Ritwika Sengupta and Daniel Trnka.

The development of the paper was also possible thanks to the support and review of other OECD colleagues from the Public Governance Directorate (GOV), in particular Andrew Blazey from the Public Management and Budget Division (PMB), as well as Science, Technology and Innovation Directorate (STI), in particular David Witikoff from the Science and Technology Policy Division (STP). It also benefited from the insights provided by the BRIDGE Expert Group and the Regulatory Policy Committee.

The paper was prepared for publication by Jennifer Stein, with editorial support from Andrea Uhrhammer.

This working paper is part of the Better Regulation in the Digital aGE (BRIDGE) Initiative, connecting conversations between policymakers, academics and technologists to provide guidance in addressing the challenges of regulating digital technologies and the use of digital tools to support regulatory processes.

Table of contents

Abstract	3
Acknowledgements	4
Abbreviations and acronyms	7
Executive summary	8
1 Introduction to the spectrum of regulatory models	9
Spectrum of regulatory models	9
Context and caveats	12
2 Guiding model choice: The contextual environment	14
Impact on innovation	14
Nature of private actors and markets	16
Public sector institutional readiness	19
Example assessment of the contextual environment	21
3 Comparative analysis of key regulatory models	24
Methodology	24
The distribution of responsibility across the spectrum	26
4 Adapting the spectrum of regulatory models to the digital economy	47
Key takeaways for the digital economy	48
The way forward	51
References	52

FIGURES

Figure 1.1. The spectrum of regulatory models	10
Figure 3.1. The spectrum of regulatory models: Specifying the role of public and private actors and levels of flexibility	27

TABLES

Table 2.1. Assessing the contextual environment to inform regulatory model choices	21
Table 3.1. Command-and-control	27
Table 3.2. Performance-based regulation	31
Table 3.3. Market-based mechanisms	34
Table 3.4. Co-regulation	37
Table 3.5. Meta self-regulation	41
Table 3.6. Industry self-regulation	43
Table 4.1. Key considerations for choosing a regulatory model	48

Abbreviations and acronyms

ACCC	Australian Competition and Consumer Commission
AI	Artificial Intelligence
AI ACT	Artificial Intelligence Act
C2PA	Coalition for Content Provenance and Authenticity
CFTC	Commodity Futures Trading Commission
CRS	Computer Reservation System
CSAM	Child Sexual Abuse Materials
DAA	Digital Advertising Alliance
DMA	Digital Markets Act
DPA	Data Protection Authority
DPO	Data Protection Officer
DSA	Digital Services Act
EIDAA	European Interactive Digital Advertising Alliance
EU	European Union
FINRA	Financial Industry Regulatory Authority
GDPR	General Data Protection Regulation
GIFCT	Global Internet Forum for Counter Terrorism
IoT	Internet of Things
LLM	Large Language Model
NFA	National Futures Association
OSA	Online Safety Act
PDPA	Personal Data Protection Act
PDPC	Personal Data Protection Commission
RIA	Regulatory impact assessment
SBA	Small Business Administration
SEC	Securities and Exchange Commission
SME	Small and Medium-sized Enterprise
SRO	Self-Regulatory Organisation
TRL	Technology Readiness Level

Executive summary

In regulatory practice, policymakers and regulators are constantly faced with the need to decide how best to address risks to social wellbeing. Their decisions depend on properly evaluating these risks and determining the appropriate degree of public intervention in balance with other considerations such as growth and productivity. To balance these considerations, policymakers must have a sound understanding of the issues at stake and the available options to make informed decisions.

In this regard, the digital economy, and innovation more broadly, are no exception. Policymakers are required to consider not only whether digital products and services *should* be governed, but *how* this should be done – specifically, whether public regulatory measures are necessary and which approaches are most proportionate and suited to the regulatory context. This challenge is amplified by the fast-evolving nature of digital products and services and the risks they pose. This dynamic environment demands a high level of technical expertise from policymakers and regulators, since inadequate protections can harm citizens' wellbeing, whereas overly stringent intervention can stifle innovation.

Given this complex and highly uncertain context, policymakers are turning to a range of regulatory responses, from testing whether existing regulatory frameworks are sufficient to deal with emerging risks, promoting standards development or voluntary industry-led approaches, to implementing new horizontal regulatory frameworks.

In this context, this paper has a dual purpose: 1) to illustrate the key facets of various regulatory models and their underlying decision criteria, and 2) to support policymakers in their decision-making processes for regulatory governance, in particular for the regulation of the digital economy. It reviews the principal regulatory models used by governments at the current juncture, including command-and-control, performance-based regulation, market-based mechanisms, meta self-regulation, industry self-regulation and a deliberate choice not to regulate. It positions them on a spectrum according to their level of rigidity versus flexibility, and the role of different actors in defining the regulatory regime and its compliance mechanisms. It then provides analysis of how regulatory policymakers are applying regulatory models to the regulation of digital activities.

The models included in the paper are established ways of describing different regulatory approaches. However, it develops a novel methodology for categorising and describing them, focusing on elements that are key for regulatory governance, in particular how regulation is developed, implemented and enforced. Public and private actors may be directly responsible for, or impact the regulatory *objectives*, *means* and *delivery* processes of a given model, with these roles varying along a spectrum. This approach allows for a more structured analysis of the models, which supports their comparison and helps to identify the trade-offs that policymakers may wish to consider when deciding on their actions, including the resulting level of flexibility, and corresponding risks and opportunities. Moreover, this analytical approach enables a more nuanced analysis of governance models, including blended models that combine elements of different theoretical models in practice.

1 Introduction to the spectrum of regulatory models

The effectiveness of regulation depends on a variety of factors, many of which lie beyond the full control of policymakers and regulators. These include: the structure of the regulated market; institutional capacity; the regulatory traditions of a given jurisdiction; the specific characteristics of a regulated area and risks it creates; as well as other environmental and cultural factors. As these elements differ across countries, there is no one-size-fits-all approach to regulate even similar or identical issues. Thus, governments often tailor regulatory responses not only to the levels and characteristics of risks, but also to their own institutional and market contexts, while also facilitating international co-operation to address transnational challenges.

This paper aims to support policymakers and regulators in navigating this complex landscape by providing insights into the decision and success criteria of different regulatory models. Its ultimate objective is to broaden understanding of the available options for governance by describing a conceptual framework, enabling more informed choices of models that align with regulatory needs, contextual factors and levels of risk.

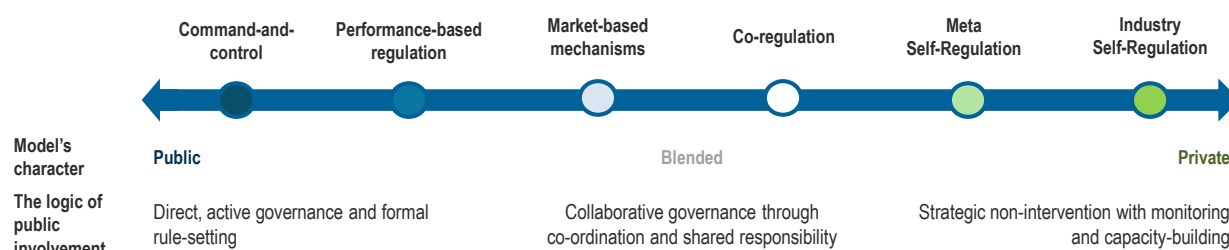
Spectrum of regulatory models

Emphasising the respective role of public and private actors as a feature to classify models, while acknowledging that both these groups are always involved in the governance process in some capacity, is supported by the literature (Verbruggen, 2013^[1]; Black, 2001^[2]; Sinclair, 1997^[3]) and reflects the changing dynamic of responsibilities within digital regulation over the past two decades (Cusumano, Gawer and Yoffie, 2021^[4]). This approach helps clarify the trade-offs policymakers and regulators face between regulatory models, considering that some models feature dominant private sector leadership, while others are heavily driven by the public sector. Layering onto this criteria consideration of the level of flexibility offered by each model to regulated entities forms the basis of the methodology for analysing regulatory models here. The resulting models that arise from this range and variation of roles can be categorised as a spectrum of regulatory models (Figure 1.1).

The spectrum comprises six models well-documented in existing research. These models are non-exhaustive, and not exclusive to the regulation of the digital economy. They are the private, public and blended regulatory models that were identified as the most commonly adopted – fully or partially. As is the premise of this paper, there exist many variations of different models based on the level of flexibility offered to the regulated entity vis à vis demonstrating compliance, and the relative levels of responsibility of public and private actors in each regulatory process. Thus, multiple additional “blended model” variations can be positioned along this spectrum, reflecting the diversity and flexibility of regulatory arrangements. As will be described in Chapter 3, the methodology used for this paper allows for more informed construction of such models. The paper does not prescribe particular regulatory models to specific digital products or activities.

Figure 1.1. The spectrum of regulatory models

Including the logic of public involvement



Source: Authors' own elaboration.

The spectrum of regulatory models is presented from the perspective of the public actors, i.e. policymakers and regulators (see Box 1.1). Models that are heavily designed and controlled by the public sector are on the left side of the spectrum, while models with lower public involvement are towards the right side. This implies that a lesser public role corresponds to higher delegation of power and responsibility to private actors (and vice versa).

Box 1.1. Public and private actors

This paper frequently uses the terms “private” and “public”, particularly to categorise actors who have a responsibility in the development or implementation of a regulatory model. For the purposes of this paper, they are understood as follows:

- **Public actors:** These are policymakers or regulators whose legitimacy stems from direct or indirect democratic mandate. In the case of the latter, they remain a part of the public administration, being staffed by public officials. This includes all branches of national governance and administration, including independent regulators, and in the case of the European Union (EU), a supra national regional regulatory body, which is included due to its unique characteristics, particularly its broad regulatory portfolio.
- **Private actors:** For the purposes of this report, private actors are taken to mean any entity subject to regulation by government. This can include private enterprise or industry, industry representative groups or organisations, and, in some cases other types of associations (for example, charities) or even individuals. The role of private actors can go beyond mere compliance, either by being integrated into the regulatory system or by their engagement in self-regulation at individual or group levels. These actors are not part of the public administration, but some regulatory power can be delegated to them.

This paper also acknowledges that regulation targeting the digital economy increasingly incorporates other non-private actors such as civil society, academia, and even individual users into regulatory frameworks. Examples such as the EU Digital Services Act (DSA) and the General Data Protection Regulation (GDPR) illustrate these evolving approaches, as will be explored further in this report.

However, in practice, regulatory approaches to digital activities often prioritise industry actors. This occurs both at a collective level – for example, through consultations or lobbying by representative stakeholder bodies – and at an individual level, for example, when the regulation is providing companies with flexibility in designing the standards and processes intended to lead to achieving the regulatory objectives. International and national standards bodies, whether public or private, often undertake broad

stakeholder engagement approaches that involve a wide range of actors in standards development, which reach across the public and private sectors and civil society.

The technical expertise held by industry actors, such as technology firms, their proprietary digital infrastructure and their role as both innovators and primary rule-takers when it comes to regulations targeting the digital economy, mean they can contribute significantly to the definition of efficient and effective regulatory regimes that achieve their goals without unduly overburdening enterprises. For these reasons, industry actors remain the main focus of the analysis conducted in this paper.

The definitions of private and public actors used in this paper have been developed specifically for its purposes and should not be treated as universal. They focus on the role of private and public actors in regulatory activities, particularly on how power is delegated within and beyond the public sphere. They are less suited to distinguishing, for example, between public and private enterprises, which – while potentially important for policy efforts – are not as directly linked to regulatory actions.

Source: Authors' own elaboration.

A regulatory model, such as command-and-control or co-regulation, for the purpose of this paper is defined by analysing the different roles of public and private actors throughout the regulatory process – relating to each model's regulatory objectives, means and delivery. In line with the OECD Recommendation on Agile Regulatory Governance, these regulatory models range in formality – from stringent binding rules to voluntary standards and an active decision not to directly engage in the governance process – and may originate from either public authorities or private entities (OECD, 2021^[5]). More details on the methodology utilised for defining and analysing the models are presented in Chapter 3.

Digital activities

The conceptual framework presented in this report is applied to the question of regulating the digital economy, as the characteristics of digital products and services make employing a single regulatory model especially challenging. Thus, regulations covering the digital economy simultaneously require an understanding of all options available to regulators and provide many examples of the implementation of various solutions. Digital technologies evolve rapidly, have cross-sectoral applications, are transboundary in their deployment and generate high levels of uncertainty. Over the past two decades, governments have seen the emergence of significant social and economic risks connected to well-established digital technologies and services, such as the challenges of information integrity (OECD, 2024^[6]) or online safety (OECD, 2024^[7]; OECD, 2024^[8]) associated with online platforms. Moreover, technological breakthroughs such as the development of Large Language Models (LLMs) in the field of Artificial Intelligence (AI) have demonstrated that the time from development to commercialisation of a technology has shortened, giving governments limited time to handle potential risks before they materialise on a large scale. In some instances, this has prompted governments to use regulations to intervene more directly – with the most significant examples being in the areas of digital platforms, online safety or fintech. However, given both capacity and technology constraints, as well as the need to balance innovation, competitiveness and regulatory goals, policymakers are increasingly opting to move away from rigid command-and-control models (at least in the first instance), instead choosing to build more flexible regulatory models that place a greater emphasis on the expertise and role of the private sector in the regulation of their technologies.

The use of these more flexible regulatory models, involving both the public and private sector's expertise and capacity, has increasingly expanded, trading off the limitations of either purely private or public regulatory approaches and acknowledging a shared regulatory responsibility (Coglianese and Nash, 2017^[9]; OECD/KDI, 2021^[10]). This finding is supported by the OECD research mapping efforts to regulate AI, which found a variety of public-private regulatory models are being used or introduced to manage the

governance challenges and opportunities of AI (Sarliève et al., 2025^[11]). The research also found that governments combined an average of two models per AI regulatory effort, with the EU, for example, using as many as four different regulatory models in the EU AI Act. These models tend to define a risk-based approach which deploys a mix of binding and non-binding regulatory requirements, while enforcement mechanisms often include shared responsibility for public and private actors.

Thus, digital products and services can serve as a useful illustration of the broader considerations related to regulatory models. For these reasons, they are used to contextualise the reflections presented in Chapter 2 on the relevant factors when choosing a regulatory model. They also provide practical examples for more in-depth analysis of regulatory models, particularly blended ones, discussed in Chapter 3.

Agile regulatory governance in evolving contexts

Based on the existing analysis of different approaches to regulatory governance, as well as looking at specific cases related specifically to digital products and services, this paper builds on four key insights:

- First, that multiple regulatory models exist, blending the roles of public and other actors in the different elements of a regulatory framework. Considering the most relevant regulatory models for a given context is increasingly important to manage the complexities of digital activities.
- Second, that this context, as well as the level of risk generated by the regulated object, can change over time, making it necessary to monitor and revise past governance choices to ensure regulation remains agile and proportionate.
- Third, that each regulatory model presents distinct risks and opportunities. Consequently, the most effective regulatory system may not employ just one regulatory model but instead combine these models, balancing their respective strengths and weaknesses. Such mixes lead to different trade-offs that policymakers and regulators should consider.
- Fourth, that effective implementation of regulatory models relies on certain conditions and always requires targeted, though varying, investment by the public (and private) actors involved. Regardless of whether a regulatory model is more public- or private-led, public actors always play a role. This means that public investments are always required in at least some elements of the regulatory system, for example for monitoring and data-gathering, even under more private-led approaches. Therefore, this paper recognises that, in some situations, a “do nothing” approach may be appropriate for a regulator. However, it defines this approach as not taking a direct role in regulating the market at a given moment. It does not imply refraining from building the capacity to understand, monitor and, if needed, engage more directly in the future, as it may be necessary for regulators to remain genuinely agile if circumstances change.

Context and caveats

This report does not suggest that models that blend the role of public and private actors are new, or applicable only to digital technologies, with existing OECD research already acknowledging these applications (OECD, 2002^[12]). Nevertheless, it goes beyond simply describing different models and formulates guidance on the relevance of, and conditions under which, each regulatory model can effectively deliver responsible, agile regulations. While the focus of the paper is on the regulation of the digital economy, the methodology created for it can also be successfully utilised for other areas, especially those characterised by similar levels of complexity and fast paced innovation.

A key limitation of the spectrum classification is its focus on the roles of public and private actors, which, although central to its purpose, overlooks factors such as how binding a model is or the specificity of its rules. In this regard, it must be underlined that the public or private character of regulatory governance does not directly correspond with the level of flexibility integrated into the regulatory framework, not its

effectiveness. For example, a well-governed self-regulatory framework can be highly effective, especially under social or regulatory pressure, as shown by self-regulatory systems for films and video games (Cusumano, Gawer and Yoffie, 2021^[4]). Similarly, public or private models are not synonymous with mandated or voluntary frameworks. For example, as will be demonstrated in Chapter 3, participation in a self-regulatory scheme supervised by the government may be legally required to access a market. To address this, the detailed descriptions of the models presented in Chapter 3, as well as the analysis in Chapter 2, include these additional characteristics where relevant. This paper does not provide an assessment as to the effectiveness of each regulatory model, as this will vary according to the regulated matter, the market and the contextual environment.

Some regulatory approaches, often presented as alternatives to regulation, are not included within the spectrum of models presented in this paper. These include, for example, information and education-based approaches, which may be important elements across all models in line with good regulation standards, or serve as the basis for self-regulatory models. Similarly, outright bans are treated here as a tool within the command-and-control model. Such approaches are considered elements of regulatory responses rather than fully developed regulatory models and therefore fall outside the framework used in this paper.

For similar reasons, the “wait-and-see” approach is not included as a defined regulatory model. It is applied when a potential regulatory issue is not sufficiently developed or understood to warrant a form of regulatory intervention (OECD/KDI, 2021^[10]; Finck, 2018^[13]; Jenik and Lauer, 2017^[14]). In such cases, public actors monitor markets to gain insights into emerging innovations, observing as they evolve into new products, services and scale. Such monitoring aims to answer two key questions related to disruptive innovations: first, identifying anticipated or emerging market risks and opportunities related to the innovation; and second, evaluating how effectively current regulatory frameworks can address these risks or enable opportunities. To achieve this, regulators undertake various activities such as horizon scanning, strategic foresight development, regulatory experimentation and stakeholder engagement, and establish observatories or fora for knowledge exchange. Therefore, regulators should not view the “wait-and-see” approach as avoiding new regulatory challenges, but rather as an intermediate period to build relevant capacities and develop proportionate regulatory action plans, which the spectrum of regulatory models can inform. It is thus better understood as part of the process of deciding whether, when and how to regulate, rather than as a regulatory model in its own right.

The definitions of the regulatory models presented are not universally termed and accepted across jurisdictions. This may lead to differences in how the strengths and weaknesses of particular approaches to regulatory governance are identified by different governments. To address this, the report draws on the most standardised and widely used definitions in the literature. In addition, by mapping the risks and opportunities of each model’s core building blocks, the analysis shows how different categories of actors shape a regulatory process, regardless of how any government labels or frames its specific setup.

Finally, two previously mentioned points require additional emphasis. First, the spectrum presented should not be seen as an exhaustive categorisation of all approaches to regulatory governance involving both public and private actors. In practice, many blended models applied in regulatory practice exist between those analysed here. Second, while the paper includes a focus on digital technologies, the strengths and weaknesses identified for each model also apply to other sectors, with digital-specific considerations being relevant for other, especially innovation-driven, areas.

2 Guiding model choice: The contextual environment

Assessing the general characteristics of the risks to be managed, private markets and regulatory institutions can support policymakers and regulators to make a more informed decision as to the appropriateness of a given regulatory model to address a regulatory need.

With regards to these elements, policymakers and regulators can ask themselves:

- What is a proportionate regulatory response to the characteristics (stage of development, risks and the nature of the associated harms) of a given product, service or activity?
- What is the nature of the private actors and markets who will be involved in and impacted by the regulatory model/s?
- Are public institutions ready, or do they need to be further equipped, to successfully develop and implement a given regulatory model?

Based on better regulation principles, along with existing OECD best practices and recommendations, this chapter presents these considerations with regards to the spectrum of regulatory models and the associated public and private roles. The considerations are presented from the perspective of policymakers, aiming to guide the public evaluation of all elements of the regulatory environment crucial for deciding on employing regulatory model. In addition, an example assessment at the end of this chapter applies these general considerations to a hypothetical digital service, to demonstrate how they can guide policymakers and regulators in choosing a more private, blended or public model of regulatory governance. Some examples from the regulation of the digital economy, together with cases from other sectors, are also used to illustrate these general considerations.

Impact on innovation

Regardless of the regulatory model adopted, regulation should start from the recognition of a longstanding interdependence between innovation and regulation. Innovative products and services introduce new challenges and risks that may require regulatory responses, while at the same time they evolve and are commercialised, and supported, within an existing regulatory environment (OECD, 2025^[15]). The key question is therefore not whether this link should exist, but rather what form it should take to foster both innovation and safe, well-functioning markets.

Public debate often oversimplifies the relationship between regulation and innovation, reducing it to the notion that stricter regulation necessarily hampers innovation (Bradford, 2024^[16]). This view overlooks other critical factors that shape innovation in any given market, such as political strategies and policies (Patanakul and Pinto, 2014^[17]), institutions and human capital (Atsu and Adams, 2023^[18]), or specifics of a local financial sector (Bradford, 2024^[16]). Most importantly, for either public, private or blended models,

certainty can encourage investment in innovation within a market, whereas uncertainty over potential future regulation can have a freezing effect (Fabrizio, 2012^[19]).

Nevertheless, concerns about the potential negative impact of regulation on innovation should not be dismissed. While there is no automatic correlation, regulation can affect innovation in two main ways. First, it may restrict innovations considered too risky or impose requirements that limit, or create path dependencies, for technological development. Second, it may create administrative burdens that constrain the broader innovation ecosystem, such as through permits, paperwork or reporting obligations (OECD, 2020^[20]). In both cases, regulatory restrictions and administrative processes can serve legitimate public interests or create unnecessary obstacles for technological development. Thus, the key is to ensure that such measures are informed by clear evidence of risk and effectiveness, and are used proportionately to the extent that intervention is justified.

Understanding the characteristics of a new product, service or business model is critical to inform the extent to which certain forms of regulatory intervention are required, and thus the relevant regulatory models for a proportional regulatory response to the challenges created, while still encouraging innovation.

Technological maturity and risk profile: A risk-based response

A risk-based regulatory response focuses on mitigating potential harms by considering different regulatory approaches based on the likelihood and magnitude of risks (OECD, 2021^[21]). It aims to ensure a proportionate level of response that avoids overly prescriptive regulation that may unnecessarily ban technologies, restrict their applications, or mandate certain standards.

Two key considerations, amongst others, in determining a proportionate risk-based response are: i) the maturity or stage of technological development; and ii) the known or anticipated level of risk posed by the innovation. Both factors should guide decisions on the appropriate degree of public and private involvement, and the overall regulatory model.

First, public actors should tailor their regulatory model to the Technological Readiness Level (TRL) of the regulated innovation. In early stages, the role of governance is often to embed innovation within an ethical framework that can guide its further development by anticipating and mitigating the risks (OECD, 2025^[15]). In the middle and later stages of development, the market begins to provide more data, such that public authorities can better assess whether anticipated risks have been addressed in the development of the technology, and if new risks have emerged (OECD, 2025^[15]).

Second, the level of risk itself should inform regulatory choices. The OECD defines risk as the likelihood of an adverse event multiplied by the severity of harm if it occurs (OECD, 2010^[22]). Where serious risks are identified, public regulatory measures may be justified even in early phases of technological development, a so-called precautionary response (OECD, 2002^[12]). A private-led, often voluntary, approach may not suffice to steer innovation away from uses deemed significantly harmful to public wellbeing, and public authorities may require stronger involvement or mandatory commitments. These actions can still be supported by industry, as in the case of Microsoft calling for public regulation of facial recognition technology (Microsoft, 2018^[23]).

Combining these two considerations, on the maturity and risk profile of a given technology, policymakers can inform the preferable level of public intervention. In early and mid-stage innovation, industry-led self-regulation – informed by public values and signals – may be most appropriate, unless the severity of risks is deemed to be sufficiently high with enough confidence. As commercialisation progresses and risks become clearer, more collaborative or formal public interventions can follow if necessary. Regulation can thus evolve alongside innovation, with different models applied as novel technologies develop.

There are a variety of agile mechanisms that governments can use to further inform the most appropriate response from a risk-based perspective. The OECD has developed a set of foundational and technology-specific values that governments can apply to help shape regulatory governance, signals that are particularly important when their involvement is limited (OECD, 2024^[24]). These values can be implemented through voluntary instruments such as codes of conduct or more innovative approaches like by-design principles, often led by industry actors at earlier stages of technological development (OECD, 2025^[15]). Regulatory experimentation can be valuable for informing the design and adjustment of publicly driven regulatory models. Different regulatory experimentation approaches can be employed across all technology readiness levels. These include policy prototyping, which can be applied throughout TRL 1–9. For less mature technologies, innovation testbeds may be used, typically at TRL 2–7. For more mature technologies, regulatory sandboxes may be more appropriate. These have typically been used at TRL 6–9 pre commercialisation, although sandboxes can be used at earlier TLR levels to provide regulatory oversight to inform future regulatory approaches, and to enhance co-operation with industry (anticipatory sandboxes). This may become increasingly relevant as regulators seek to understand emerging technologies (OECD, 2025^[15]). In all these forms, regulatory experimentation can help bring regulators and innovators together to reinforce the link between innovation and regulation, ensuring that innovation continues while risks are properly managed under an appropriate model.

Nature of private actors and markets

The nature of private actors and markets, alongside the relationship of these with public actors, are key considerations across all of the regulatory models discussed in Chapter 3. Involving stakeholders throughout the regulatory process is a core principle of better regulation and supports more effective, informed outcomes (OECD, 2012^[25]). Moving beyond traditional consultation, involving private actors more directly in different stages of the regulatory process can also help build trust between regulators and industry (Cusumano, Gawer and Yoffie, 2021^[4]; Albareda, 2008^[26]).

Private actor willingness and capacity for regulatory governance

With the exception of command-and-control, all regulatory models presented in Chapter 3 require private actors to play an active role in shaping regulatory objectives, means for compliance, or delivery. In this context, two key questions should be considered when selecting a model:

- Are private actors genuinely willing to assume responsibility for parts of the regulatory process in support of public, rather than private, goals?
- Do private actors have the necessary capacity to meet these responsibilities and achieve compliance?

These factors should be assessed not only at the outset – for example when evaluating regulatory alternatives to public models in an regulatory impact assessment (RIA) (OECD, 2020^[27]), but also through ongoing monitoring, as private actors' motivations and capacities can evolve over time. This is crucial, as Chapter 3 illustrates how commitment to voluntary regulatory frameworks may be short-lived.

Assessing these factors in practice can be challenging. Evaluating willingness is complex, as clear criteria are difficult to define. Generally, willingness to engage in self- or co-regulation is strongest when broader public and private policy objectives are aligned, especially when regulation is in the self-interest of the industry (OECD, 2015^[28]; Cusumano, Gawer and Yoffie, 2021^[4]; Beard and Nye, 2011^[29]; King and Lenox, 2000^[30]). One motivating factor is the ability to use regulation to enhance the public perception of safety, making products more appealing to consumers. This incentive may be even stronger in the context of existential risks, as in the case of a nuclear sector (Sinclair, 1997^[31]). Similar motivations are present in the digital sector, where leading tech firms have expressed support for regulation in areas like AI and privacy

to help maintain consumer trust (Bradford, 2024^[16]). However, policymakers should also be mindful that regulatory engagement by industry may sometimes serve private aims not aligned with public goals. For instance, it can be used to mislead consumers about product safety by presenting it as regulated, thus safer, a concern raised by the EU regarding cryptocurrencies (Howcroft, 2025^[31]), or to raise entry barriers that favour dominant firms (Macedoni and Weinberger, 2024^[32]).

Nevertheless, public actors still have some tools to assess the willingness of the industry to regulate. This can become evident in the process of stakeholder engagement, and in the propositions of regulation formulated by private actors. Moreover, to help gauge private actors' genuine willingness for effective regulatory self-governance, policymakers and regulators can look to past behaviour. This can offer insights into the likelihood of industry compliance with more privately led regulatory models, such as voluntary frameworks. A poor history of adherence may suggest similar outcomes in future efforts (Short and Toffel, 2010^[33]).

The level of trust between private and public actors is also crucial in this regard. Delegating more responsibility for regulatory management to the market itself – either by allowing industry to self-regulate or by engaging it under more blended models – requires a sufficient degree of confidence between both parties. For public actors, this provides greater certainty regarding the motivation and capacity of industry. For private actors, it gives confidence that self-regulation will be fairly evaluated, and that industry will have a genuine chance to address the risks it generates, with the state intervening in good faith only when required. Such trust can naturally grow through successful industry-led or collaborative efforts over time.

With regards to private actor capacity, different models place different capacity demands on industry. Performance-based and market-based approaches require some level of independent policy expertise, enabling companies to design internal measures that meet regulatory goals. Models such as industry self-regulation, meta self-regulation and co-regulation demand even greater capacity, particularly for active involvement in delivery, which may require generating detailed reports, self-assessment or even ensuring compliance by collective monitoring and enforcement, e.g., through a Self-regulation Organisation (SRO).

Importantly, public actors can actively support and protect private actor capacity in areas especially relevant for regulatory governance. Under the General Data Protection Regulation (GDPR), for instance, some categories of entities are required to designate a Data Protection Officer (DPO). DPOs play a key role as a contact point for the organisation, responsible for advising it, but also monitoring or documenting data processing. Although they remain part of the regulated organisation, including private companies, their role in the governance system is supported by specific safeguards. The GDPR provides that they must have adequate resources, cannot be dismissed for performing their duties, and must report directly to senior management (EU, 2016^[34]). The European Commission has also issued guidelines to support DPOs, including those embedded within private organisations (EU, 2017^[35]).

Where willingness and capacity of private actors is high, policymakers and regulators can have more confidence in the effectiveness of models that rely more heavily on private actor leadership, especially if the level of confidence between both groups is high. However, as previously noted, both the willingness and capacity of the private sector to engage in self- or co-regulation should be monitored over time. This may take the form of strategic monitoring and dialogue. The Australian Competition and Consumer Commission's (ACCC) Digital Platform Services Inquiry exemplifies this approach. Since 2017, the ACCC has been scrutinising digital platforms, delivering 14 reports and 33 recommendations addressing both public and private actors, including specific companies, facilitated by public submissions and ongoing dialogue between the regulator and companies regarding their practices (ACCC, 2025^[36]). This not only allows for timely adjustments to the regulatory model in cases of underperformance, but also applies regulatory pressure. Continued oversight can encourage the industry to maintain effective private or collaborative frameworks in order to avoid more direct public intervention (Cusumano, Gawer and Yoffie, 2021^[4]; Rees, 1997^[37]), with the industry efforts taking place “in the shadow of the public regulator” (Newman and Bach, 2004^[38]).

Determining market characteristics

The assessment of market characteristics is critical to ensure that private actors are capable of creating an effective, collective system of regulatory governance. A more collaborative approach may be necessary to enable public actors to compensate for some of the shortcomings detected, for example, by taking a more active role in co-ordinating the self-regulatory efforts or enforcing regulation originated in the private sector.

The level of self-organisation (Miracle and Nevett, 1988^[39]; OECD, 2015^[28]) and homogeneity of the market are important factors to be evaluated (King and Lenox, 2000^[30]; OECD, 2015^[28]; Cusumano, Gawer and Yoffie, 2021^[41]). Well organised markets, in which companies regularly co-operate using more formal measures, e.g., through industry organisations or by maintaining informal contacts, lower the barriers to industry alignment, co-ordination and enforcement, as it is easier to build consensus and detect potential wrongful activities (OECD, 2015^[28]). This can increase the scope for effective self-regulatory models, accounting for the other variables that might impact the relevance of these models. Further, the total number of firms in the industry itself can also prove crucial, with free-riders in self-regulation easier to detect in smaller markets (Walters and Wiseman, 2022^[40]).

The size of the companies in the market is also a key consideration, as smaller companies can have limited resources to engage in regulatory activities (Sinclair, 1997^[3]). There can be a tension in the optimal level of regulatory flexibility suited to such firms. For example, under performance-based regulations small companies may benefit from increased flexibility in designing how they comply with regulatory requirements to suit their current processes and resources. At the same time, this flexibility could increase burdens should sufficient guidance on compliance not be provided, or if this approach requires greater resourcing and capacity than a firm can provide. Moreover, smaller companies can be marginalised compared to big resourceful companies in co-regulatory efforts if not sufficiently included in the process. This is crucial for digital markets, in which start-ups often play an important role.

Integrity, transparency and accountability

Public actors should also take measures to ensure that regulatory systems remain resilient to risks such as regulatory capture and conflicts of interest, which threaten the integrity, transparency and accountability of regulation (OECD, 2014^[41]; OECD, 2017^[42]). Specifically considering the level of market concentration, market homogeneity, industry self-organisation and size of market actors (OECD, 2017^[42]).

Capture can take various forms depending on how much influence public or private actors have over a regulation's design and implementation. Thus, it can be more challenging to identify capture in regulatory models where private actors are more involved in the regulatory processes.

Regulatory models towards the centre and private-led end of the spectrum, such as co-regulation and meta self-regulation, are especially vulnerable to integrity risks linked to the proximity between regulators and the regulated. These risks apply to both public and private regulatory bodies. As with other forms of stakeholder engagement, proximity in itself is not problematic, issues tend to arise more when one group dominates the process (OECD, 2017^[42]).

Careful management is required to ensure equal access to policymakers and regulators and avoid capture. While there are standard mechanisms for incorporating diverse private input (OECD, 2025^[43]), they can still be misused. In a 2013 OECD survey, nearly half of both lobbyists and regulators acknowledged that inappropriate influence-peddling was at least an occasional problem (OECD, 2017^[42]). Moreover, even when interactions are procedurally appropriate, more resourceful stakeholders may have greater access, creating risks of imbalance (OECD, 2017^[42]).

In models such as command-and-control or performance-based regulation, particular attention should be given to maintaining integrity in lobbying and public consultation, as these are key channels for stakeholder influence. This includes ensuring that all relevant parties are represented, including smaller or less prominent actors in the industry (OECD, 2017^[42]). In co-regulatory models, transparency becomes even more critical due to shared responsibility in designing the means for complying with the rules (OECD, 2017^[42]).

There are various strategies that public actors can employ to strengthen integrity, transparency and accountability in their relationship with private actors and the OECD has issued guidance to strengthen the integrity of this relationship. This includes establishing legal and institutional frameworks for interactions, ensuring transparency and accountability, and securing equal access for all stakeholders (OECD, 2024^[44]; OECD, 2003^[45]; OECD, 2017^[42]).

Digital tools are also complementary to efforts aimed at increasing integrity, transparency and accountability. In market-based models for example, AI tools can help detect irregularities, such as anomalies in public fund allocation (OECD, 2024^[46]). (OECD, 2021^[47]). More broadly, digital tools can also help assess and simplify regulation, reducing the risk of regulatory complexity becoming a barrier to market access for smaller firms (OECD, 2024^[46]).

Across all models, it is important that regulators foster an organisational culture that supports transparency and integrity. Officials should receive guidance and training on identifying risks and applying mitigation measures. Rules should also be in place to manage conflicts of interest, including those arising from future employment (OECD, 2003^[45]; OECD, 2024^[44]).

Public sector institutional readiness

Regulator skills and capacity should be considered as a factor when choosing a regulatory model. For example, a command-and-control approach may be considered the most suitable model to manage severe risks. However, such a model would be ineffective if the public actors did not have the institutional skills to appropriately identify and assess risks, to propose appropriate mitigations, or to monitor them and enforce compliance. In such cases a co-regulatory model may be more practical to better manage risks by sharing responsibilities between public and private actors. This approach could help overcome information asymmetries between the regulated entity and the regulator. Hence, different regulatory models call for distinct skills and expertise, digital tools and institutional setups to be most effective in governing new technologies.

Expertise and skills

Sufficient expertise and skills are foundational to effective regulatory governance across all stages of the policy cycle. Choosing more collaborative or industry-led models should not be seen as a way to outsource responsibility or avoid public investment in expertise. At a minimum, public actors must understand how a regulated product or service works, how its risks emerge, and what mechanisms are realistically available to manage them. While collaboration with industry is valuable, public bodies should also invest in their in-house capabilities to avoid over dependency and ensure independent oversight.

That said, the level and nature of expertise required can vary depending on the regulatory model. In command-and-control models, where regulators define the means of compliance, technical expertise must be deep to ensure that regulatory design does not impose unnecessary compliance burdens or onerous compliance mechanisms. In performance-based models, however, regulators need not prescribe the precise compliance methods but must understand the range of viable options and explain these to the market as necessary, letting the market determine the means most appropriate for themselves. Investments in regulator expertise may be prioritised towards market monitoring and surveillance of real-

world implications of the regulatory framework, and linking these back to where risk and value emerged along the value chain.

While the fundamental level of understanding of a given technology should be always built by public actors, they do not always need to have the capacity to conduct specific processes, such as AI audits, themselves. However, where such activities are not undertaken within the public sector, the relevant regulator would need the expertise and capacity to judge whether industry-led audits are robust or if further intervention is warranted. Thus, foundational technical knowledge should remain a priority under any regulatory model. In this context, regulators may consider criteria for balancing their own knowledge investments and oversight arrangements, with using external expertise. Skills necessary to maintain the key elements of a regulatory model may require investments in internal capacities. Without it, regulators may struggle to design effective frameworks, evaluate compliance, or shape industry-led initiatives.

In addition, integrity and transparency measures are crucial to ensure proper relationships between the administration and external actors, in line with OECD recommendations and data (OECD, 2024^[44]). More specific measures, relevant for particular models, are discussed in Chapter 3.

Digital tools

Institutions need to be equipped to effectively enforce the regulatory governance systems put in place. As such, the required sophistication of digital tools is often determined by the role assigned to a public actor within a given regulatory model. In command-and-control models, for instance, AI systems can support more effective inspections (OECD, 2025^[43]). In more flexible regulatory frameworks, digital tools can help regulators guide compliance and evaluate diverse approaches to meeting regulatory objectives. Under performance-based regulation, where regulators may lack capacity to review every compliance case and where firms may be uncertain about how to use the available flexibility, digital self-assessment systems can ease the burden on both sides. One such example is a virtual assistant for GDPR compliance, developed jointly by data protection authorities and universities in Croatia and Italy (ARC2, 2025^[48]).

A further example of a cross-functional digital tool is Estonia's X-Road data exchange platform, which created shared infrastructure for inter-agency data sharing. It enables public institutions such as the police or tax authority to access data from their respective registries, while citizens only need to provide information once, in line with the "once-only principle". This makes the regulatory system more efficient in its use of information and reduces reliance on fragmented or ad hoc solutions. (E-Estonia, 2025^[49]). Such platforms are key to establishing data-driven regulatory goals and informed strategies, whether for direct public regulation or shaping private governance.

As a minimum, policymakers and regulators should maintain situational awareness of digital developments that affect regulatory needs; this is critical even when a government is waiting to assess its options. Where governments is the leader on a given process, it should strive to have effective digital tools that can help effectively manage it— noting that this does not necessarily equate to the latest, most advanced tool. For regulatory institutions, it is important that digital tools are tried, tested and understood to an extent where the tool's integrity can be assured.

Institutional co-ordination

Some regulatory models may heighten co-ordination challenges and require greater institutional investment. As discussed in Chapter 3, performance-based regulation and other flexible models often define regulatory objectives in broad terms. This can lead to two issues: "mission creep", where regulators exceed their intended mandate (OECD, 2014^[41]), and difficulty in interpreting objectives, which may result in weak enforcement or inadequate protection. Transparency and accountability mechanisms can help mitigate these risks. Likewise, regular engagement with other public actors, private actors and other associated stakeholders can reduce the risks of misalignment (DRCF, 2025^[50]).

In more collaborative models, it is particularly important that all stakeholders understand their respective roles and responsibilities, as well as those for others involved in the regulatory model (OECD, 2014^[41]). This is crucial in models such as co-regulation and meta self-regulation. In co-regulation, a lack of clarity can lead to inconsistent interpretations between public and private actors. In meta self-regulation, where public oversight is central, clearly defined mandates are vital for private-public co-operation for regulatory enforcement. These functions should be explicitly set out in legislation (OECD, 2014^[41]). For example, the US Securities Exchange Act of 1934 (U.S., 1934^[51]) clearly defines the responsibilities of self-regulatory organisations. One-stop-shops can also contribute to this goal by enhancing visibility over institutional responsibility and integrating systems among involved agencies (OECD, 2020^[20]; Askim et al., 2011^[52]).

Finally, in models like command-and-control, performance-based or market-based regulation, where public actors lead on delivery, a co-ordinated enforcement strategy is especially important. Effective enforcement supports the broader functioning of these models and, in the case of performance-based and market-based regulation, helps avoid fragmentation by applying consistent categories across diverse regulatory mechanisms. In this context, enforcement systems themselves should be co-ordinated and consolidated, to prevent duplication and overlap (OECD, 2014^[53]). Integrated information and communication technologies, such as Estonia's X-Road, can be instrumental in supporting this co-ordination and improving enforcement efficiency (OECD, 2014^[53]).

Overall, there are benefits to all regulatory models in enhancing the institutional setting for co-ordination. Formal or informal networks of regulators offer the most basic means of co-ordination, serving as platforms for building shared understanding, forming joint strategies or simply acting as repositories of relevant information (OECD, 2025^[43]). These networks, however, can often face diminishing returns based on the required time commitment required by participants. Thus, consideration should be given to the required commitment of time, resources and money vs the extent of required co-ordination. Sophisticated system integration for data sharing can often be a complementary solution, that builds in collaboration without drawing heavily on individual time but requiring investment in the digital tools highlighted in the previous sub-chapter.

Example assessment of the contextual environment

Table 2.1 is an example of the assessment and decision-making process a policymaker or regulator could employ to inform the choice of regulatory model/s based on the contextual environment. This is just one way of implementing the considerations offered here as a demonstration of how this paper's considerations can be utilised. For this exercise, a hypothetical digital service is used, with its characteristics developed specifically for its purpose.

Table 2.1. Assessing the contextual environment to inform regulatory model choices

A hypothetical case of a digital service

Step	Characteristics	Outcome
Impact on innovation		
Assess the level of technological maturity	The assessed service remains in a dynamic growth phase. Although many companies now offer products integrating the same technology, new applications continue to emerge and technological breakthroughs are still being made, attracting fresh entrants with evolving business models.	Growth (TRL 6): requiring flexibility for the service to continue to evolve
Assess the risk level	Different applications for the service exist. These applications have varying levels of risks, both on individual, sectorial and social level. There are several applications deemed to be medium – high risk.	Mostly high and medium risks. Elements of more public models may be necessary to manage some of the more serious risks, including mandatory standards preventing the use of assessed service for dangerous or illegal activity.

Step	Characteristics	Outcome
Nature of private actors and markets		
Assess the capacity and willingness of private actors	There are many mature companies developing the service that have been involved in self-regulatory schemes in the past. However, the effectiveness of these self-regulatory schemes has not been proven, and the industry remains divided on the type and level of governance required for the service. The market is also highly concentrated, with many large actors. These actors have sufficient capacity for regulatory governance. However, smaller actors involved in the development of the service, or its downstream applications, may not have as many resources for governance and regulatory compliance.	Potentially weak incentives for self-regulation based on historical evidence of self-regulatory commitment by some companies. Also, a varying level of capacity for self- or regulatory compliance across the industry. A more public approach to developing regulatory objectives and enforcement may therefore be required, drawing on mechanisms such as performance-based regulation and meta self-regulation.
Assess the market characteristics	Is the market rather homogenous? No. A diverse range of companies offering the service Is the market power concentrated? Yes. Development of the service, as well as the technology on top of which it is built is dominated by a few major companies, limiting but not preventing new entrants Is the industry self-organised? No. Existing industry organisations focus on sharing information in specific areas; intense competition inhibits broader co-operation and the establishment of collective incentives for governance. Are there many small actors in the market? Yes. While leading service providers possess resources for self-regulation, numerous smaller actors may require flexibility in how they comply with regulations,	Elements of co-regulation, performance-based and market-based incentives may be the most appropriate, rather than self-regulation. Public-led solutions are probably necessary in areas where there is limited willingness/lack of common understanding of problems by the industry.
Assess the effectiveness of Integrity, transparency and accountability measures	Public actors already have mechanisms for ensuring the integrity of lobbying and engagement. However, there is a lack of extensive experience in public actors working with blended models of regulations and thus ensuring the integrity of process remains when engaging in stronger public-private collaborative models. This may result in limited process transparency if there are not appropriate management strategies deployed when there are significant shared roles between public and private actors in the regulatory model.	The regulatory system is equipped with mechanisms sufficient for ensuring integrity of consultation and lobbying process crucial for more public models. New guidelines and processes should be developed for ensuring the integrity of the collaboration with the private actors in enforcing the regulation and monitoring the market (as blended models are likely to be employed).
Public institutional readiness		
Assess availability of digital tools	Regulators have existing tools for risk-based assessment and enforcement that can be leveraged for this digital service. However, these tools may need to be adapted or expanded to keep up with the demands placed on it. Data-sharing and joint testing between regulators has been increasing, however there are still improvements required for the effective use of digital tools, especially given the cross-sector and cross-regulatory nature of the service.	Pending nature of regulatory models employed, further investment in digital tools may be required particularly for monitoring high-risk areas.
Assess existing skills and expertise	Mostly sufficient in addressing existing high-level risks – suggesting that it is useful to maintain and/or reinforce elements of current regulations. Insufficient skills and expertise regarding some new high- and medium-level risks – suggesting introducing elements of collaborative models to promote knowledge sharing.	Public actors are generally well equipped to address harmful interactions with the service in the areas already covered by existing digital and non-digital regulations. However, the volume of risks may be expanding, requiring more resources to manage the implementation and enforcement of regulations. Public actors have a less mature understanding of new risks provoked by the service. Developing this understanding requires close collaboration with private actors, as well as further skill development on how the service functions and is used.

Step	Characteristics	Outcome
Assess levels of institutional co-ordination	The assessed service cuts across mandates. Clear role allocation, joined-up strategies, and routine collaboration mechanisms are not mature or consistent within government, limiting consistent regulatory enforcement where new approaches are required.	Consider models that can leverage components of private-led enforcement
Proceed with the identified model(s)		
Proceed with the identified model/elements of models (examples of possible regulatory responses within identified models)	High-risk areas may require stronger or reinforced command-and-control measures. This can be most applicable where such approaches or regulations are used to already manage existing risks, given there will be a strong public institutional understanding of issues and likely existing digital tools for monitoring and enforcement.	Elements of command-and-control
	For more evolving, new risks there are greater unknowns and less public actor expertise on the management of these risks. The private sector likely has a stronger understanding and experience on building technological standards for governance. That said, a role for public oversight to manage variability in incentives and capabilities across private actors is required. Creating consortia with the industry for monitoring new risks and co-developing standards may be required.	Elements of co-regulation
	To complement the development of regulatory governance with industry, public actors can use performance-based regulation to set clear goals and principles that should guide industry in creating practices and technical solutions that embed governance requirements. This provides flexibility for private actors and can allow for differences in capacity between larger and smaller actors.	Elements of performance-based regulation
	Given the growth stage of technological development, market-based mechanisms can be used as a complementary measure to the above regulatory models to embed regulatory objectives within further technological developments.	Elements of market-based mechanisms

Source: Authors' own elaboration.

3

Comparative analysis of key regulatory models

Regulatory models are specific combinations of regulatory design and delivery choices. Each of them features a distinct governance structure that assigns specific functions to these actors. This chapter demonstrates that all regulatory models come with unavoidable trade-offs and implementation challenges related directly to the impact of private or public actors. It analyses each model by breaking them down into “building blocks,” three key processes in the regulatory cycle. It then assesses the degree of responsibility and accountability public and private actors hold for each process, while identifying the resulting level of flexibility available to comply with the regulatory regime.

Methodology

Each model carries specific risks, opportunities and prerequisites, depending on how responsibility is distributed across the processes. By linking these findings to individual building blocks rather than to entire models, the analysis remains relevant even when policymakers combine elements from different approaches. For example, a tailored, blended model could use performance-based regulation to set regulatory objectives and for regulatory means, and meta self-regulation for enforcement. This report can better inform decision-making by applying process-specific insights to each element of such blended models for policymakers and regulators.

Key processes in the regulatory cycle

To break down the responsibilities of different actors across the regulatory models, this report considers three key processes in the regulatory cycle. These are: 1) defining regulatory objectives, 2) regulatory means and 3) regulatory delivery. They should not be treated as the exhaustive representation of the whole regulatory policy cycle. However, they remain aligned with the general OECD definition, which includes identifying a public policy issue, regulatory means, delivery and review (OECD, 2012^[25]; OECD, 2025^[43]; OECD, 2025^[54]).

The categories used in this report were adapted to better reflect the building blocks that distinguish different governance models. They were identified through an analysis of various approaches to regulatory governance, many of which are presented in this paper. For this reason, some categories were broadened. In particular, regulatory delivery also includes elements related to implementation and review processes.

Some others, namely regulatory objectives and regulatory means, are defined following the logic presented by C. Coglianese’s classification of regulatory tools, focusing on how they refer to the *ends* and *means* of the regulation accordingly (OECD, 2010^[22]). This approach was motivated by the fact that, depending on the model, particular aspects of the regulatory regime may or may not be specified directly in the legal instrument that establishes the regulations. For example, with performance-based regulation, the regulatory objectives (the ends) will be mentioned in the regulation in the form of required outcomes or

principles, while regulatory means, or the processes and mechanisms that must be put in place to achieve those outcomes, defined by private actors, naturally will not be set out in the legislative text. The definitions adopted in this paper make it possible to identify such provisions in legal or regulatory documents and analyse the governance model they create. This is exemplified by the excerpts from specific regulatory acts after each definition below:

- **Regulatory objectives:** these refer to the targets or goals to be achieved or avoided, specifying what *results* should be realised. Crucially, regulatory objectives as defined in this paper are not synonymous with policy objectives, which define the overarching, strategic direction for any regulatory activity in an area. Regulatory objectives operationalise policy objectives in the specific regulatory acts, including specific outcomes or more general principles (OECD, 2010^[22]). For example: *Industry participants take reasonable and proactive steps to prevent access or exposure to, distribution of, and online storage of class 1A material* (Australia, 2023^[55])
- **Regulatory means:** these are specific actions or processes intended to achieve the regulatory objectives, detailing *how* the targets should be met. When translated into regulation they can take the form of a detailed procedure or standard. For example: [...] *the categories or classes of radio equipment referred to in point 1 of this Part shall: be equipped with the USB Type-C receptacle, as described in the standard EN IEC 62680-1-3:2021* (EU, 2022^[56]).
- **Regulatory delivery:** these include all activities aimed at promoting compliance and achieving regulatory objectives. Thus, it should be understood broadly, covering activities such as analysis, inspections, licensing and permitting, guidance and prevention, data-collection and monitoring or market surveillance, but also enforcement actions in a narrower sense, such as sanctions (OECD, 2018^[57]). When translated into regulatory frameworks they can describe enforcement mechanisms, such as inspections or testing, but also the instruments for monitoring or overseeing the efforts of regulated entities. For example: [...] *the Commission may impose on the provider of the very large online platform or of the very large online search engine concerned fines not exceeding 6 % of its total worldwide annual turnover in the preceding financial year [...]* (EU, 2022^[58]).

The role of public and private actors

Public or private actors can exert different levels of control over the key processes constituting the regulatory models. To categorise the degree of control, this paper employs an adapted version of the RACI matrix, an established management technique for allocating responsibility among actors involved in the process (Cabanillas, Resinas and Ruiz-Cortés, 2012^[59]).

The modification of the matrix comes from the acknowledgement of the specific nature of allocating responsibility across the regulatory cycle. First, it should be noted that accountability to government is a distinct aspect of a regulatory regime and will be present unless a regulatory regime is entirely voluntary and industry-led. When responsibility is delegated, as in the example of performance-based regulation, where responsibility for deciding *how* to achieve the outcomes may be left to the regulated entity, accountability to government will still remain as a key requirement of the regulations. For this reason, a category of *accountable* stakeholder was excluded, as it is assumed that accountability always lies with the public actors, aside from the case of industry-self regulation. Second, the categories of *consulted* and *informed* stakeholder were merged, as in the analysed context, they played the same role, namely involving a non-responsible group of actors in the development of the process. The only difference comes from the nature of the actor responsible for the process. In the case of the public sector leadership, it is assumed that private actors should always be consulted, not merely informed, in line with good regulatory practices. The same cannot be assumed for private-led processes, which are often not subject to similar requirements and standards.

Thus, two categories are utilised to describe the role public and private actors can play in specific processes constituting the governance model:

- **Responsible:** a group of actors in charge of developing and implementing the element of the regulatory framework in question.
- **Consulted or informed:** a group of actors given a chance to formulate their expectations toward the process and trying to ensure their interests are reflected in the approach taken by the responsible actors.

Flexibility under different models

This paper recognises the limitations of defining regulatory models solely by the roles of public and private actors and therefore describes them also according to the level of flexibility granted to regulated entities. This perspective complements the actor-based approach and provides an additional analytical lens.

Flexibility is assessed at the level of the model's building blocks, reflecting the degree of freedom available in determining the elements of regulatory frameworks outlined above. This enables a more nuanced analysis of different models. As the descriptions of specific models illustrate, the level of flexibility generally corresponds to the model's position on the spectrum. However, greater or lesser flexibility in specific processes may also arise under more public or more private models respectively, depending on how roles are distributed across the overall model.

For example, although performance-based regulation can be categorised as closer to the public end of the spectrum, it assigns responsibility for regulatory means (determining the process by which the regulated can demonstrate sincere efforts to achieve target outcomes) to private actors, which may provide regulated entities with greater flexibility. In some cases, the general logic of the model also needs to be considered. Under market-based mechanisms, for instance, public actors are responsible for regulatory delivery, but participation is not always mandatory. As a result, potential regulated entities may benefit from a higher degree of flexibility with regard to regulatory delivery.

The distribution of responsibility across the spectrum

Figure 3.1 specifies the respective roles of public and private actors in the three key regulatory processes analysed, and the relative level of flexibility granted to regulated entities over each aspect of the regulatory model. As previously highlighted, attempting to analyse the effectiveness or stringency of these models would not be meaningful here, as this will depend on the regulatory objectives, the actors, the market, the design of the regulatory regime itself and other contextual factors. In the later part of the chapter, the models are described using the criteria defined above: the roles of public and private actors, and the level of flexibility afforded to regulated entities. Each model is also accompanied by a brief case study illustrating how it can be applied to the regulation of the digital economy.

Figure 3.1. The spectrum of regulatory models: Specifying the role of public and private actors and levels of flexibility

Specifying the role of public and private actors and the level of flexibility allowed

Process	Actors	Command-and-control	Performance-based regulation	Market-based mechanisms	Co-regulation	Meta Self-Regulation	Industry Self-Regulation
							
Regulatory objectives	Public	Responsible	Responsible	Responsible	Responsible	Responsible	Consulted or Informed
	Private	Consulted or Informed	Consulted or Informed	Consulted or Informed	Responsible	Consulted or informed	Responsible
	Flexibility	Low	Low	Low	Medium	Medium	High
Regulatory means	Public	Responsible	Consulted or Informed	Consulted or Informed	Responsible	Consulted or Informed	Consulted or Informed
	Private	Consulted or Informed	Responsible	Responsible	Responsible	Responsible	Responsible
	Flexibility	Low	High	High	Medium	High	High
Regulatory delivery	Public	Responsible	Responsible	Responsible	Responsible	Responsible	Consulted or Informed
	Private	Consulted or Informed	Consulted or Informed	Consulted or Informed	Responsible	Responsible	Responsible
	Flexibility	Low	Low	Medium	Medium	Medium	High

Source: Authors' own elaboration.

Command-and-control

Command-and-control is one of the oldest and most traditional forms of public regulation, often treated as its synonym. It relies on a clear hierarchy and role distribution (Hodges, 2016^[60]) where public authorities set prescriptive rules and oversee their implementation by entities under their jurisdiction (OECD/KDI, 2021^[10]). That requires governments to monitor compliance, normally via specialised regulatory authorities with responsibilities to conduct inspections and enforcement activities (Kamara, 2017^[61]).

Table 3.1. Command-and-control

The role of private and public actors over regulatory processes, resulting flexibility, risks and opportunities

Regulatory process	Role		Flexibility	Risks	Opportunities
	Public actors	Private actors			
Regulatory objectives	Responsible	Consulted or informed	Low	Rigid goals may stifle innovation.	Addressing serious risks
				Overly unilateral regulation not considering interests of all stakeholders	Supporting innovation through increased clarity and ensuring it will follow a safe path set out by common standards.
Regulatory means	Responsible	Consulted or informed	Low	Lack of agility leading to regulatory lag	Can facilitate coherent, holistic vision and long-term harmonisation of the public interests with innovation
				Global fragmentation of technology and governance	Stabilising the market, providing legal clarity, especially in high-risk areas
Regulatory delivery	Responsible	Consulted or informed	Low	Overly unilateral delivery, based mostly on strict control	Bolstering harmonisation and consumer's standing in the market
				Insufficient positive incentives for compliance and lower legitimacy in the eyes of industry	More secure and harmonised approach to sensitive public concerns
					Greater legitimacy in the eyes of citizens and stronger transparency

Source: Authors' own elaboration.

As the most public-driven regulatory model, public actors are responsible for all three regulatory processes (see Table 3.1). However, in practice, private actors, mainly the industry, can still play a significant role, particularly during the creation of regulations and the management of their outcomes, making a purely public model of regulation more of a theoretical concept (Sinclair, 1997^[3]). First, private actors participate in consultation processes and lobby to influence regulatory objectives. Second, they create non-binding instruments, such as standards or codes of conduct, which policymakers should and do consider when crafting regulations. Finally, private actors can affect regulatory delivery through formal advisory roles with agencies or by voluntary informal actions like sharing or withholding supportive resources such as data. Nevertheless, such allocation of roles significantly limits the flexibility of regulated entities in all aspects of regulatory governance.

Command-and-control faces three key limitations. First, it is often criticised for stifling innovation (Benoliel, 2004^[62]; Wahyuningtyas, 2019^[63]). The rigidity of strict regulations or highly prescribed regulations creates a risk of disconnecting the rules from fast-evolving dynamic realities, as they may be too inflexible (Benoliel, 2004^[62]; OECD/KDI, 2021^[10]; OECD, 2024^[6]). It may also limit flexibility and opportunities for private entities to develop potentially more tailored, cost-effective compliance methods (Coglianese, 2016^[64]).

Second, mandatory rules and standards can lead to the fragmentation of global governance if other jurisdictions do not adopt similar measures, undermining the universal nature of many services and products and disadvantaging users. This is well exemplified by the internet, as its universal character can be easily compromised by national or regional rules and standards imposed, impacting its technical and governance layer (Drake, Cerf and Kleinwachter, 2016^[65]; Perarnaud et al., 2022^[66]). This underlines the importance of international regulatory co-operation, which can ensure the maintenance of established standards in constructing regulatory responses.

Finally, the model is criticised for being seen as overly unilateral, offering few incentives for compliance in today's complex regulatory landscape, where sanctions alone can be insufficient to motivate industry for compliance (Hodges, 2016^[60]; Black, 2001^[2]) and requiring significant investment in properly robust compliance systems by industry, and enforcement systems by public actors (Sinclair, 1997^[3]).

Despite its challenges, the command-and-control regulatory model can still prove effective in digital regulation when used in the right contexts, and with sufficient investments. It is particularly relevant for addressing substantial risks to key values that require traditional protection measures such as mandatory standards or bans (OECD, 2002^[12]). As such, it can be employed not only to safeguard citizens but also to clarify the permissible scope for innovation and align it with long-term policy objectives, as demonstrated in banning social scoring systems under the EU AI Act (EU, 2024^[67]). The broader impact of command-and-control on innovation also remains more complex than it is often presented (Malloy, 2010^[68]), as inconclusive empirical research on environmental regulation in China demonstrates (Yi et al., 2019^[69]; Sun et al., 2021^[70]; Li et al., 2018^[71]), and as was discussed in Chapter 2 in regard to public regulation in general. In scenarios involving significant risks and limited regulatory options (OECD, 2002^[12]), detailed public commands can provide necessary clarity, legal certainty and stability for the market to encourage innovation (European Commission, 2023^[72]).

Command-and-control also benefits from its long-standing presence in regulatory practice. In some regulatory cultures (Senden et al., 2015^[73]), countries with extensive experience in this model may achieve better outcomes by leveraging existing expertise rather than adopting unfamiliar approaches (OECD, 2002^[12]; European Commission, 2023^[72]). Moreover, the model has evolved through initiatives like regulatory impact assessments (RIA) and stakeholder engagement, reducing its unilateral nature (OECD, 2002^[12]). These efforts, which the OECD actively champions (OECD, 2020^[27]; OECD, 2014^[53]), have improved the model's effectiveness and adaptability, by creating more informed practices that involve a wider range of actors at all stages of the regulatory process.

Applying the model for digital products and services

Command-and-control is seldom an approach adopted for governing digital services. The risk of potentially stifling innovation remains a significant concern for many governments, which are reluctant to apply the most public regulatory model to fast-evolving sectors driven by rapid technological advancement.

Imposing and enforcing a specific standard on digital companies is also limited in practice due to public actors' capacity constraints. Regulations often integrate existing private digital obligations, or solutions such as technological standards (McAllister, 2014^[74]; Griffin and Vander Maelen, 2023^[75]). Box 3.1 presents the reliance on private standards even when implementing the classical version of the command-and-control model to fast-evolving technologies underlying digital services.

Box 3.1. The EU “common charging” solution

Imposing USB-C and USB Power Delivery as mandatory standards

The EU's changes to the Radio Equipment Directive and the Ecodesign Regulation are examples of traditional command-and-control regulation applied to new technologies. These changes mandate the inclusion of one industry standard, namely USB-C receptacles, in all portable devices covered by the directive, as well as the standardisation of cables, chargers, and the Power Delivery protocol for devices supporting fast charging.

While presented as beneficial for both the internal market and citizens, the directive clearly prioritises consumer cost, convenience, and positive environmental impact over any negative effects on manufacturers or innovators, which is acknowledged in the accompanying impact assessment study. Flexibility is introduced to the approach through the ease of making future amendments, as the Commission will monitor impacts on innovation and, if needed, update the list of regulated devices and required technologies. Additionally, manufacturers can still equip regulated devices with additional receptacles of different standards, as long as they include the mandated standard as well.

The example of common charging solutions illustrates the difficulty of imposing prescriptive rules that are universally acceptable and beneficial. It also highlights the tension between mandatory standards and innovation as well as balancing the interests of certain businesses versus consumers.

Source: (European Union, 2023^[76]) (European Union, 2022^[77]) (European Union, 2019^[78]) (European Commission: Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs, Economisti Associati, Fraunhofer, IPSOS, Trinomics and Vazquez, Y., 2021^[79]).

The potential for governance fragmentation mentioned above is an important risk, however, command-and-control can conversely be a useful model to prevent commercial fragmentation. This may be particularly relevant for some digital services, for example not allowing for the creation of “walled gardens”, where service providers enforce varying obligations on their users via their respective Terms of Service (Drake, Cerf and Kleinwachter, 2016^[65]). Command-and-control can also serve as an effective measure to support innovation by harmonising rules or standards when the activities of companies deploying technology may limit certain uses or access to digital technologies, i.e. consumer “lock-in”, as demonstrated in Box 3.1.

Other risks related to public actors being solely responsible for regulatory delivery can be also magnified in the case of digital services. While these issues are very context-dependent and should not be considered as set in stone (Sinclair, 1997^[3]; Cole and Grossman, 2018^[80]), in the regulation of digital products and services they can provoke some challenges. For example, cross-sectoral and data-driven areas like

privacy require even more costly mechanisms for monitoring and inspections (Kamara, 2017^[61]), and are further strained by resource and expertise shortages (OECD, 2022^[81]).

While acknowledging the complexity of the relationship between innovation and regulation, as described in Chapter 2, there is a tension between imposing mandatory standards to combat commercial fragmentation and addressing risks, and competitiveness. The EU's Digital Rulebook illustrates this, as it is unclear whether its digital standards will drive global adoption (Bradford, 2020^[82]) or increase isolation of the EU's digital market (Komaitis, 2023^[83]). Companies such as Apple and OpenAI, for instance, have already delayed or withdrawn product launches in Europe, citing regulation as the main reason (Chee, 2024^[84]; Davies, 2024^[85]).

For these reasons, the use of this model remains limited, with its elements often employed in the form of banning certain practices, rather than enforcing specific standards. This is, for example, one of the approaches used under the EU AI Act, outlawing certain uses of high-risk AI systems (EU, 2024^[67]). Nevertheless, even when dealing with substantial risks, policymakers often opt to use performance-based or co-regulatory models, as will be demonstrated below.

Performance-based regulation

The underlying logic of performance-based regulation is that public actors define the regulatory goals that regulated entities must achieve without prescribing rigid means for how they should do so (OECD, 2002^[12]). These goals can be formulated to be precise and measurable, as in the outcome-based regulation approach (OECD, 2021^[21]), or rely on formulating the broader, general rules that should guide private actors, as in principle-based regulation (Black, Hopper and Band, 2007^[86]). Despite the level of detail used to define the expected results, the key element of the model is allowing industry to develop its own methods for compliance.

Performance-based regulation is expected to result in an allocation of roles that reflects market dynamics and innovation. It allows public actors to define proportionate regulatory objectives based on the assessment of risk to safeguard citizens' well-being, while allocating significant flexibility to private actors to use their expertise to determine the means that deliver safer products and services. The approach to regulatory enforcement is equally important, as studies on the model's application highlight the need for sufficient public monitoring and enforcement to ensure that the intended outcomes are actually achieved (OECD, 2021^[21]; OECD/KDI, 2021^[10]; Coglianese, 2016^[64]).

Despite its clear advantages, the model should not be treated as a silver bullet for all regulatory challenges. First, it still requires substantial expertise from public actors in defining regulatory objectives. Overly broad objectives can create lack of clarity about what constitutes compliance, leading to challenges for regulatory monitoring (Coglianese, 2016^[64]) and inconsistencies in interpretation between regulators and regulated entities, particularly across different jurisdictions (Black, Hopper and Band, 2007^[86]). This hinders international regulatory co-operation and makes compliance more difficult, while also opening new avenues for regulatory arbitrage, with companies taking advantage of gaps or differences in regulations or laws (Pollman, 2019^[87]). The ultimate effect can be insufficient protection of citizens under the regulation.

Table 3.2. Performance-based regulation

The role of private and public actors over regulatory processes, resulting flexibility, risks and opportunities

Regulatory process	Role		Flexibility	Risks	Opportunities
	Public actors	Private actors			
Regulatory objectives	Responsible	Consulted or informed	Low	Vague objectives risk inconsistent application and regulatory arbitrage, as well as “mission creep” by regulators	Supports a risk-based, technology-neutral framework that enhances legitimacy in the eyes of regulated entities Allows for building a flexible regulation evolving together with innovation
Regulatory means	Consulted or informed	Responsible	High	Flexibility can create additional burdens for smaller companies without sufficient resources and policy expertise	Balancing the opportunities of flexibility with guidance and objectives introducing stability Allows for regulatory innovation Alignment of the processes and standards with business models leading to higher compliance
Regulatory delivery	Responsible	Consulted or informed	Low	High capacity required from public actors to ensure market surveillance, enforce the standards and processes developed by industry Need for developing strong capacity to guide and co-ordinate industry's efforts	Using regulatory enforcement to ensure consistent interpretation of the regulation

Source: Authors' own elaboration.

A further risk of this model is that overly broad objectives can lead to “mission creep”, with regulators overstepping their initial mandate, by creating uncertainty about the actual extent of the regulation (OECD, 2014^[41]). This can complicate not only the enforcement of the regulation but can also hinder collaboration efforts between different sectoral regulators, unsure about their respective role in the regulatory system. This is especially important in a cross-sectoral reality for digital products and services.

Finally, public actors should be ready to reimagine their approach toward supervision and enforcement (Black, Hopper and Band, 2007^[86]). When applied to smaller companies without extensive legal expertise, such as technology start-ups, the responsibility to develop their own compliance methods can impose additional burdens (Coglianese, 2016^[64]; Balleisen and Eisner, 2009^[88]) and, in turn, hinder innovation. Even in the case of creating detailed, measurable outcomes, regulated entities should not be expected to possess all knowledge and resources required to develop the processes and standards required to achieve the objectives set by the state (OECD, 2021^[21]). Thus, regulators have a role in providing industry with additional guidelines to help regulated entities achieve compliance, as suggested by the OECD Best Practices (OECD, 2014^[53]).

With sufficient investment in these areas, performance-based regulation can prove particularly effective in ensuring compliance while creating agility and enabling a flexible, adaptive regulatory framework that keeps pace with innovation, while ensuring long-term focus on identified risks. The model allows for developing industry-driven solutions that can better fit company business models, integrating compliance into internal processes rather than imposing external mandates. This can foster greater internalisation of both the compliance mechanisms and the regulatory objectives within the structures of the regulated entities (OECD, 2021^[21]; Black, Hopper and Band, 2007^[86]).

Applying the model for digital products and services

Performance-based regulation is one of the most popular solutions applied to the regulatory management of digital products and services. This is due to two main reasons. First, it allows for a high level of flexibility by setting the objectives aligned with the risks provoked by a service, that can guide its design and adaptation throughout its whole lifecycle. When designed in a technologically neutral manner, performance-based regulation should not require frequent modification, as broad objectives can continue to consistently guide industry towards compliance (World Economic Forum, 2020^[89]). This can be further supported by employing risk-based approaches, creating different objectives for different regulated entities under a single regulatory framework, aligned with the level of risk they generate, as demonstrated by the EU DSA and Digital Markets Act (DMA) or the UK Online Safety Act (OSA).

Second, it can help public actors to benefit from industry's technical expertise. It can even provide them with the space to innovate not only in technology but also in compliance methods, developing new and more efficient means of meeting regulatory ends with active support from regulators. For instance, the Coalition for Content Provenance and Authenticity (C2PA) technical standard allows for tracing content origin, showing how private solutions can help achieve public goals, in this case related to countering misinformation and disinformation (Coalition for Content Provenance and Authenticity, 2025^[90]). Nevertheless, the same flexibility naturally gives policymakers and regulators less control over the compliance process, as presented in Box 3.2.

Box 3.2. Australian Social Media Minimum Age Act

Performance-based approach to regulating access to digital services

In 2024, the Australian Parliament adopted changes to the Australian Online Safety Act. The new rules entered into force in December 2025 and require online service providers categorised as “age restricted social media platforms” to take reasonable steps to prevent children under 16 from creating or holding an account, under the threat of fine. This approach leaves private actors with significant flexibility in determining how to comply, although some limits apply to data processing and the use of government issued identification for age verification, which is allowed only as an alternative method.

The novelty of the approach means that its effectiveness cannot yet be fully assessed, although several potential risks can already be identified. Platforms must first self-assess whether they fall within the scope of an age restricted service, which initially created some lack of clarity regarding the scope of the regulation. In addition, available age verification tools vary considerably in effectiveness. Combined with the way the objective is defined – focusing on limiting account creation rather than access to the platform itself – these factors may reduce the overall effectiveness of the Act.

Regulators have taken several steps to mitigate these risks and support compliance. The eSafety Commissioner, one of the two authorities responsible for enforcing the Act alongside the data protection agency, has published extensive guidance for regulated entities, including a self-assessment tool for platforms. In addition, the Government conducted a study evaluating the effectiveness of different age verification methods and the Minister for Communications has issued a legislative instrument clarifying which services fall outside the scope of the Act. The eSafety Commissioner also consulted with major social media companies prior to the Act's entry into force to help ensure that the largest platforms were prepared for the new requirements. The regulator maintains and regularly updates a list of services considered to be within scope.

These measures do not fully address the limitations associated with current age verification technologies or the possibility of accessing services without creating an account. However, they provide meaningful support to entities seeking to comply with the regulation. More broadly, the case illustrates how enforcement approaches may need to adapt under performance-based regulation, as regulators still need to remain active and capable to aid regulated entities in utilising the flexibility delegated to them.

Source: (Australia, 2024^[91]; Australia, 2025^[92]; eSafety Commissioner, n.d.^[93]; Australia, 2025^[94]; eSafety Commissioner, n.d.^[95]).

Applying performance-based regulation to digital services creates some additional challenges. When applied to this area, performance-based regulation more often takes the form of principle-based regulation rather than outcome-based regulation. This is to a high degree because it can be difficult to clearly define measurable outcomes, such as safety and security impacts, for digital technologies and associated products, as opposed to physical technologies such as drones (OECD, 2024^[96]). This can accelerate the risks related to regulatory objectives, specified in Table 3.2, resulting from lack of clear understanding of the objectives.

The experience of the Personal Data Protection Act (PDPA) in Singapore illustrates how such broad principle-based approaches may require additional guidance to ensure consistent interpretation. The PDPA is largely composed of general desired outcomes rather than prescriptive rules. For example, Article 12a states that “an organisation must develop and implement policies and practices that are necessary for the organisation to meet the obligations of the organisation under this Act” (Singapore, 2012^[97]). Moreover, the PDPA allows for processing data in the legitimate interest of the processing entity, which, while often used in privacy regulations, can lead to substantial interpretational challenges (Kramcsák, 2023^[98]). To reduce this risk, the Personal Data Protection Commission issued advisory guidelines clarifying key concepts and providing tools to help organisations interpret these principles more consistently (PDPA, 2013^[99]).

Market-based mechanisms

The market-based mechanisms model utilises market incentives to encourage industries to act in ways that achieve protection from societal or environmental risks, without imposing rigid rules or processes in the shape of legally binding regulations (Stavins, 2000^[100]). Thus, while not regulation in the strictest sense, market-based incentives play an important role in broader regulatory governance as captured in this paper.

Most common examples of market-based mechanisms with a clear regulatory nature are mechanisms such as licensing and permit schemes, used for regulating market behaviour by setting a cap on undesirable activities (Stiglitz, Joseph. E; Rosengard, 2015^[101]). Taxes are also an example of a market-based mechanisms, although these can serve to both disincentivise or encourage action towards a regulatory objective (managing risks), but can also be purely motivated by the need to collect revenue. (Stiglitz, Joseph. E; Rosengard, 2015^[101]). Public procurement rules and processes can also serve to nudge market behaviour in a particular direction, as well as more direct supply side policy interventions, which however do not have such a clear regulatory character and are voluntary in nature for participants.

Thus, there are similar characteristics between market-based mechanisms and other regulatory models including:

- Objective-oriented constraints: They are designed to achieve public policy outcomes (e.g., emission reductions) through obligated behavior (paying, holding permits).
- Non-compliance (e.g., failing to pay a tax or surrender permits) triggers penalties; thus, they rest on the same legal authority as mandatory standards.

- Institutional oversight: Public agencies set parameters, monitor markets, and adjust rules (e.g., in spectrum auctions and management).

Despite its reliance on market incentives, in practice, the model depends heavily on public actors and their perception of risk. Governments define the objectives that markets are expected to deliver and oversee functions related to enforcement (Mazaheri et al., 2022^[102]). At the same time, industries may retain nearly total freedom in selecting the means for reaching the targets, thus the close alignment with performance-based regulation mentioned earlier (Fisher, 2010^[103]). In regard to regulatory delivery, companies can benefit from a higher degree of flexibility than suggested by the role allocation alone. The model allows them to choose not to alter their behaviour at all, instead opting to absorb the financial costs of continuing discouraged practices, whether through higher taxes or limited access to government subsidies for example (Stavis, 1990^[104]).

One of the most widely used market-based mechanisms for regulatory purposes is the marketable permits system. Originating in environmental regulation, it sets a cap on undesirable activities, such as emissions, and allocates permits to companies. Firms that reduce their emissions below their allowance can trade surplus permits, effectively creating a market for emissions reduction (Stavis, 1990^[104]). In this way market mechanisms do not merely discourage undesirable market practices, they can also develop positive incentives to achieve desired outcomes of public policy. These mechanisms are essential for many new technologies, for example electromobility. Market-based mechanisms also proved successful for regulating access to the market and fostering development of regulated sectors with the use of spectrum auctions in the communication sector (see Box 3.3).

Table 3.3. Market-based mechanisms

The role of private and public actors over regulatory processes, resulting flexibility, risks and opportunities

Stage of the regulatory process	Role		Flexibility	Risks	Opportunities
	Public actors	Private actors			
Regulatory objectives	Responsible	Consulted or informed	Low	Demands significant knowledge about market realities to predict the impact of the objectives and address the identified risks	Aligning market processes with strategic public policy goals
Regulatory means	Consulted or informed	Responsible	High	Flexibility creating additional burdens for smaller companies without sufficient resources and policy expertise	Balancing the opportunities of flexibility with guidance and objectives introducing stability Enhancing regulatory innovation Alignment of the regulatory means with business models leading to higher compliance
Regulatory delivery	Responsible	Consulted or informed	Medium	Potential for misallocating funds without sufficient understanding of the logic of the market and technology	Significantly reducing the costs of delivery

Source: Authors' own elaboration.

The flexibility that defines the market-based approach is both its greatest strength and its biggest risk. In practice, this model represents a sophisticated form of regulation whose success hinges on careful design, particularly in terms of oversight mechanisms. First, public authorities should recognise that market-based incentives affect different types of businesses in different ways. For example, OECD data suggests that tax incentives have a greater impact on R&D efforts in small and medium-sized enterprises (SMEs) than in large corporations, partly due to the lower initial investment requirements for SMEs (OECD, 2023^[105]). Furthermore, subsidies are more likely to drive actual innovation in line with regulatory objectives when

applied during the early stages of research, which is both the most expensive phase and the period when firms focus primarily on innovation rather than commercialisation (Mazaheri et al., 2022^[102]).

Second, a major risk lies in the specific design of market-based mechanisms. OECD data indicates that direct government investments have the greatest impact within the first two years, after which firms that rely on private funding tend to outperform them (Berger, Dechezleprêtre and Fadic, 2024^[106]). Additionally, without adequate monitoring and oversight, governments risk misallocating public funds to initiatives that do not align with desired regulatory goals and do not address risks sufficiently (Li et al., 2018^[71]; Ren, Sun and Zhang, 2021^[107]).

Crucially, when well-designed and properly overseen, market-based incentives can be highly effective in fostering innovation aligned with public policy objectives while remaining significantly more cost-efficient than rule-based regulatory models (OECD, 2002^[12]; Mazaheri et al., 2022^[102]). A key advantage of this approach is that by leveraging market forces to implement policy, public authorities can reduce many costs associated with traditional regulation. At the same time, they retain control over strategic priorities, ensuring that innovation in market mechanisms aligns with the broader public interest. This is particularly relevant in the context of government investments, where public actors can support the growth of companies that may be deemed too risky or too small to attract private funding, while tying innovation to directional safety requirements for example (OECD, 2023^[105]).

Applying the model for digital products and services

Market based mechanisms remain a popular option for governing digital services and can follow several approaches within the model described above. One approach focuses on addressing specific risks, for example those related to competition in the market. Similar to traditional permit-based approaches used in environmental regulation, policymakers may create a market for a particular resource required by a service, such as a type of infrastructure. At the same time, the market can be designed to help achieve specific regulatory objectives. Frequency auctions, illustrated in Box 3.3, provide a clear example of this approach.

Box 3.3. The 5G Auction in Chile

Market-based mechanisms for an effective regulation of limited resources

Spectrum auctions are a common market-based mechanism used by regulators to manage the communications sector and encourage innovation. The importance of 5G networks for consumers and industries, for example in developing the Internet of Things (IoT) infrastructure, has made auctions a popular tool for quickly regulating the use of technology and enabling the necessary infrastructure. Their frequent use comes with variations in design, allowing regulators to pursue different objectives and adapt solutions to the local context. The Chilean spectrum auction of 2021 offers a good illustration.

Chile opted for a first-price package auction, in which companies bid for a set of blocks from a specific frequency band rather than for individual blocks and submitted their offers without knowing the prices offered by competitors. The bands were auctioned consecutively from low to high frequencies, with only one block available for each of the two low bands and fifteen in the high band (divided into five in the lower subsection and ten in the higher subsection). Spectrum caps limited the share of a band that any one company could use. In addition, mandatory coverage and bandwidth requirements were applied.

This approach accelerated the process and ensured that successful bidders, including new entrants, secured enough blocks to operate and provide adequate coverage. The auction also generated significant revenue for the government.

The case shows how regulatory means can evolve in response to local conditions. Chile's previous 4G spectrum allocation followed a "beauty contest" model, with three companies competing for three blocks based on technical proposals, and a cap of one block per company. This reduced competition and made it difficult for regulators to verify individual commitments. By contrast, the 2021 auction introduced changes that embedded regulatory objectives in the auction design, including coverage and bandwidth obligations, pro-competition measures such as caps, and a transparent bidding process.

Source: (Escobar et al., 2023^[108]; Levin and Skrzypacz, 2016^[109]; Milgrom, 2004^[110]).

Market based mechanisms are also used to support particular types of digital services or business models through supply-side policy interventions that, while not regulatory in the strict sense, can complement and reinforce market-based regulatory approaches in order to achieve regulatory objectives. This can occur either through direct support or through more indirect market shaping measures. A direct example is the America's Seed Fund Initiative in the US. The programme connects small businesses with federal agencies that provide free equity funding for selected projects and technical support through the US Small Business Administration (SBA) managing the framework (US, n.d.^[111]). It also supports the commercialisation of funded products, either by facilitating their entry to the market or by enabling sales to federal agencies through the programme's marketplace (US, n.d.^[111]).

A more indirect approach based on public procurement can be illustrated by the G-Cloud programme in the United Kingdom. Since 2014, successive iterations have allowed public authorities to purchase cloud computing services, including hosting, software and support, through a digital procurement marketplace with verified vendors (UK, n.d.^[112]). Similar to the America's Seed Fund Initiative, the programme also aims to increase the visibility of small and medium sized enterprises and make their services more accessible to the public sector. In 2019, 90% of suppliers on the framework were SMEs (UK, n.d.^[113]) and in 2023-2024 they accounted for 44% of total spending (Think Digital Partners, 2025^[114]).

Both programmes share a clear common goal: fostering the commercialisation of digital solutions by connecting them with potential public-sector users, while creating an environment that is particularly favourable to SMEs. While these examples are better understood as innovation and industrial policy tools rather than regulatory mechanisms per se, as they do not directly manage risks or create compliance obligations, they illustrate how public actors can shape the market conditions that underpin effective regulatory governance. By broadening the supplier base and supporting SME participation, such interventions can help address the market concentration and capacity imbalances discussed in Chapter 2, which in turn affect the viability and effectiveness of regulatory models applied to these markets. Moreover, public procurement frameworks can serve as vehicles for embedding regulatory objectives, for example, by conditioning market access on compliance with specific security, privacy or interoperability requirements, thereby blurring the boundary between market-shaping policy and regulation.

However, the design of such schemes remains crucial, as illustrated by the Chilean case mentioned above. In addition, supply side interventions may face challenges in defining clear criteria for selecting companies to support. For example, many government programmes prioritise funding companies that are already innovative and capable of securing private financing (Cuerva, Triguero-Cano and Córcoles, 2014^[115]; Jové-Llopis and Segarra-Blasco, 2018^[116]). For these reasons, supporting and connecting digital service provider with potential public buyer, as in the UK and US case, may be an interesting alternative to avoid challenges related to direct funding.

Co-regulation

Co-regulation remains one of the most challenging models to define, often described simply as existing between command-and-control and self-regulation (Senden et al., 2015^[73]; Sinclair, 1997^[3]). Its overarching rationale can be understood broadly or more specifically. The broader definition considers co-regulation as collaboration between public and private actors, either by dividing responsibility for different parts of regulation or by co-developing it in its entirety (Hirsch, 2010^[117]). This approach treats all blended models such as performance-based regulation and meta self-regulation as co-regulation, (Cusumano, Gawer and Yoffie, 2021^[4]) which can limit analytical depth. Therefore, this report adopts a more specific understanding of co-regulation, as a model implying co-creation of the whole regulation by private and public actors (Verbruggen, 2009^[118]).

Crucially, shared responsibility is not necessarily equal for each regulatory process, and collaboration can take different forms. For example, public actors may allow private entities to supplement primary regulation with specific instruments such as voluntary standards or codes (Griffin and Vander Maelen, 2023^[75]). This differentiates co-regulation from models such as performance-based regulation, where public actors are not actively designing standards and processes required by the regulation, or meta self-regulation, where private actors are not actively enforcing them. Instead, co-regulation creates a complex, yet coherent regulatory system that avoids control over a given stage of the regulatory process being concentrated with a single category of actors (see Table 3.4). Thus, co-regulation should be viewed as a framework integrating diverse regulatory tools and approaches.

Table 3.4. Co-regulation

The role of private and public actors over regulatory processes, resulting flexibility, risks and opportunities

Stage of the regulatory process	Role		Flexibility	Risks	Opportunities
	Public actors	Private actors			
Regulatory objectives	Responsible	Responsible	Medium	Ambiguous, consensual goals can cause misalignment between public and private actors Potential problems with transparency which can undermine public trust	Leverages diverse technological and policy expertise to align interests and create more agile and realistic goals, based on common understanding of risks and innovation. Creates more chances for including a wide range of stakeholders not limited to industry
Regulatory means	Responsible	Responsible	Medium	Different priorities for producing public and private standards and rules Reducing fundamental concerns to technical problems Opacity of the process leading to its domination by the most influential actors	Harnessing private experience and enhanced public legitimacy Better integration of existing private and public standards or rules Public actors harmonizing industry's efforts and preventing fragmentation
Regulatory delivery	Responsible	Responsible	Medium	Regulatory capture Problems with transparency and accountability Complexity of the governing structures requiring high capacity from both public and private actors	Benefiting from industry's operational expertise Enhancing compliance through greater industry ownership Reducing the administrative burden on public authorities Can provide a framework for responsible involvement of civil society in the process

Source: Authors' own elaboration.

The success of co-regulation is limited by two key risks, amongst others: conflicts of interest; and capacity of public and private actors to regulate. Like self-regulatory schemes, co-regulation depends on the alignment of public and private interests (Cantero Gamito and Marsden, 2024^[119]), in this case relevant for all regulatory processes. Especially in high-risk areas, policymakers should carefully assess the threat of misaligned priorities. In such cases, when industry cannot effectively self-police, a lack of transparency and oversight can lead to symbolic compliance (Gellman and Dixon, 2011^[120]). A lack of transparency can also empower dominant stakeholders to exert a disproportionate influence, compromising the public interest and potentially, weakening compliance incentives.

This second challenge is capacity. Even with industry goodwill, effective collaboration requires adequate resources from the industry (Balleisen and Eisner, 2009^[88]). The complexity of co-regulation can be challenging for policymakers and regulators without prior experience in such close public-private co-operation. This, in turn, exacerbates the legitimacy and accountability risks mentioned earlier.

Despite these challenges, co-regulation remains a highly effective way to align public and private regulatory efforts. It can be used to integrate existing public and private solutions, by incorporating existing industry-led initiatives into the more public solutions. (McAllister, 2014^[74]; Griffin and Vander Maelen, 2023^[75]). Conversely, when private actors take the lead in regulatory co-creation, they can implement existing public rules into the frameworks they develop (Walters and Wiseman, 2022^[40]).

Co-regulation also helps public actors to leverage industry expertise, reducing information asymmetry and aiding development of more responsive and agile regulatory frameworks (Cantero Gamito and Marsden, 2024^[119]). Its flexibility allows also for greater citizen participation at various regulatory levels, as will be demonstrated below.

Applying the model for digital products and services

Co-regulation can remain an attractive alternative to more public approaches for reasons similar to other blended regulatory models. It allows policymakers to benefit from industry expertise, while its flexible structure may mitigate some of the risks associated with delegating full responsibility for regulatory processes to private actors.

For this reason, co-regulation may enable a more efficient distribution of roles across the regulatory process compared to models such as performance-based regulation. Public actors can define overarching regulatory principles, which are then refined by industry or broader stakeholder groups. This approach is illustrated by the EU Strengthened Code of Practice on Disinformation and the Code of Conduct on Countering Illegal Hate Speech Online+. In the case of the former, public actors outlined the general measures expected from industry in the DSA, which serves as the broader framework for the Code, and in guidelines issued by the European Commission (EU, n.d.^[121]). These principles were then translated into more detailed measures by an Assembly, created for this purpose in 2021 (EU, 2021^[122]). The Assembly included previous signatories, digital service and advertising companies, and representatives of civil society, and was chaired by a representative of the Commission overseeing the process.

A similar approach was adopted for the Australian Online Safety Act (AU OSA), under which the eSafety Commissioner can require industry to develop codes translating broad principles into specific outcomes, standards and processes (Australia, 2021^[123]; eSafety Commissioner, 2025^[124]; eSafety Commissioner, 2025^[125]).

Co-regulation can also affect the delivery of regulation. The GDPR provides an example through Data Protection Officers (DPOs), who are embedded within private organisations whose core activities involve systematic data processing or the handling of sensitive or large-scale categories of data (EU, 2016^[34]). Although located within private entities, DPOs play an important role in regulatory enforcement by advising on GDPR obligations, monitoring compliance and liaising with public data protection authorities and citizens.

The EU DSA also incorporates private actors into the delivery of the regulatory framework through mechanisms such as transparency reporting and audits. These instruments require some platforms to regularly assess and disclose information about their activities, e.g., publishing yearly reports on content moderation (EU, 2022^[58]). In other cases, the regulation relies on assessments carried out by external private actors. Article 37 of the Act requires some platforms to be subject to yearly audits by independent auditors, who review the platforms' compliance (EU, 2022^[58]).

Applying co-regulation to digital services can also illustrate the potential to involve a broader range of private actors in regulatory processes. For example, the DSA framework creates opportunities for organisations, including civil society actors, to participate in content moderation. Once confirmed by national competent authorities, these entities may contribute to regulatory delivery either as trusted flaggers responsible for identifying potentially illegal content or as out-of-court dispute settlement bodies (EU, 2022^[58]).

More broadly, the EU DSA illustrates how a regulatory framework can create space for different regulatory tools and stakeholders, including elements of co-regulation. While companies are often granted flexibility in designing compliance mechanisms, the framework encourages adherence to standards and codes developed by industry either independently or in collaboration with public actors (see Box 3.4). It may also illustrate how voluntary codes and standards adopted while a technology or market is still developing can later contribute to the development of more coherent and potentially mandatory regulatory frameworks.

At the same time, digital services also illustrate some of the broader risks associated with co-regulation. For example, in the case of the EU Code of Practice on Disinformation, the European Commission and signatory companies assessed compliance differently, focusing on distinct risks in their respective reports (Borz et al., 2024^[126]), which may suggest varying interpretations and priorities regarding the regulation. Similarly, the proceedings of the Assembly working on the Strengthened Code of Practice on Disinformation were confidential. While this may have been necessary for productive discussions within a broad group, it could but could still undermine public trust in the regulatory process (Marchant, Tournas and Gutierrez, 2020^[127]), which illustrates the importance of transparency in co-regulatory processes.

Box 3.4. The Codes of Conduct under the EU DSA

Integrating voluntary commitments into a co-regulatory framework

Article 45 of the EU DSA allows for the creation of voluntary Codes of Conduct, which may be officially recognised by the European Board for Digital Services (EBDS) as potential means of demonstrating compliance with the provisions of the Act. These codes may also include additional provisions beyond the obligations specified in the DSA, provided that certain criteria are met. In particular, they are expected to define specific objectives and key performance indicators, which need to take account of the needs and interests of all relevant stakeholders. Whether the codes meet these conditions is assessed by the European Commission and the EBDS, which also monitor and evaluate progress in implementing the Act.

The same Article also indicates that the Commission is expected to facilitate the development of voluntary codes of practice. In doing so, it may invite multiple stakeholders to participate in the drafting process, including regulated entities, relevant public authorities and civil society organisations.

In addition, the DSA contains more specific provisions relating to three areas: online advertising, accessibility and crisis situations. The development of additional codes and protocols in these fields is particularly encouraged, and the Act outlines the minimum objectives that such initiatives are expected to address.

In 2025, the EU Strengthened Code of Practice on Disinformation and the Code of Conduct on Countering Illegal Hate Speech Online+ were integrated into the DSA framework. Both instruments had been created before the adoption of the DSA and were later adapted through the co-regulatory process to better align with its objectives. This illustrates how the co-regulatory framework can support the harmonisation of existing arrangements that were previously positioned more towards the private end of the regulatory governance spectrum with the broader regulatory framework.

Source: (EU, 2026^[128]; EU, 2025^[129]; EU, 2025^[130]; EU, 2022^[58]).

Finally, while the integration of tools such as standards into the regulatory frameworks can be useful for regulating digital services, overreliance on this tool and its use in areas traditionally regulated by legislation can provoke new problems (Benoliel, 2004^[131]). One of the most important ones is the risk of reducing important issues such as fundamental rights to a mere technical problems and lack of transparency compared to the traditional legislative process (Cantero Gamito and Marsden, 2024^[132]; Gornet and Maxwell, 2024^[133]).

Meta self-regulation

Meta self-regulation is another model characterised by complex interactions between public and private actors, with both acting as regulators. Meta self-regulation should be understood as a *mandated* form of self-regulation (Gunningham and Rees, 1997^[134]) in which private actors regulate themselves under the direct oversight and pressure of public authorities (Cusumano, Gawer and Yoffie, 2021^[4]; Parker, 2002^[135]). While related to industry self-regulation, it is not synonymous with it (Coglianese and Mendelson, 2010^[136]) and shares many traits with co-regulation. In contrast to industry self-regulation, which operates through instruments and structures influenced by – but independent of – government, meta-self-regulation involves direct supervision of self-regulated industry by public regulators. Hence, while the industry continues to regulate itself, it must adhere to the organisational structures and principles set by the state. Nevertheless, public actors often tend to define regulatory objectives as general guiding principles rather than specific outputs, leaving industry with considerable flexibility in how these are operationalised.

One of meta self-regulation's most crucial elements is industry co-ordination, which is essential for effective enforcement across the sector. Meta self-regulation often relies on Self-Regulatory Organisations (SROs), composed of industry members, to develop and enforce rules and standards. Membership in SROs is often mandated by law. In other words, SROs are responsible for the regulatory delivery and enforcement but are also held to account by public actors. The United States offers multiple examples of such frameworks, particularly in financial regulation, as well as in the nuclear and energy sectors (Walters and Wiseman, 2022^[40]). This is why the concept of meta-regulation often refers to frameworks in which “regulators regulate regulators” (Parker, 2002^[135]).

This overarching self-regulatory logic significantly influences public involvement in the model. Public actors are not required to define detailed outcomes or specific risks (Coglianese and Mendelson, 2010^[136]). Instead, they set guiding principles or outline the general processes that industries should adopt, overseeing their implementation and often mandating certain mechanisms or standards – such as safeguards to prevent the most influential members from capturing the SRO.

Table 3.5. Meta self-regulation

The role of private and public actors over regulatory processes, resulting flexibility, risks and opportunities

Stage of the regulatory process	Role		Flexibility	Risks	Opportunities
	Public actors	Private actors			
Regulatory objectives	Responsible	Consulted or informed	Medium	Lack of co-ordination and problems with including all relevant actors leading to dominance of the companies with the highest market power	Ensures that the outcomes designed by the industry are actually delivered.
Regulatory means	Consulted or informed	Responsible	High	Flexibility creating additional burdens for smaller companies without sufficient resources and policy expertise	Balancing the opportunities of flexibility with guidance and objectives introducing stability Encouraging regulatory innovation Alignment of the regulatory means with business models leading to higher compliance
Regulatory delivery	Responsible	Responsible	Medium	Regulatory capture Lack of established SROs in many digital sectors who can be effectively integrated into the governance structure and inconsistent implementation A chance of SROs realising industry's interests as opposed to regulatory objectives Lack of clearly defined risks can hinder regulators' opportunities to ensure the regulation is enforced in a way protecting citizens' wellbeing Problems with transparency and accountability	Benefiting from industry's operational expertise Enhancing compliance through greater industry ownership Reducing the administrative burden on public authorities

Source: Authors' own elaboration.

The main risks of meta self-regulation stem from industry co-ordination and its relationship with public actors. First, effective meta self-regulation depends on a well-functioning SRO (Miracle and Nevett, 1988^[39]; Cusumano, Gawer and Yoffie, 2021^[4]). The history of advertising self-regulation in the UK and the US suggests that successful self-regulation requires the involvement of all relevant stakeholders, such as advertisers, publishers, and broadcasters, co-ordinated by an umbrella organisation (Miracle and Nevett, 1988^[39]; Beard and Nye, 2011^[29]). However, such co-ordination can be difficult, particularly in larger industries, and enforcement may be challenging. This is why the model tends to be more effective in smaller, tightly connected industries where detecting and addressing non-compliance is easier (Walters and Wiseman, 2022^[40]).

A further risk arises from competition between SROs. If multiple bodies regulate the same industry, fragmentation can occur. This can be mitigated by making SRO membership mandatory. For example, in the US, two key SROs – the Financial Industry Regulatory Authority (FINRA) and the National Futures Association (NFA) – oversee the securities and derivatives industries, respectively, under the supervision of the Securities and Exchange Commission (SEC) and the Commodity Futures Trading Commission (CFTC). With few exceptions, participation in these self-regulatory schemes is mandatory for firms engaged in relevant activities, reinforcing their effectiveness.

Finally, public oversight is essential to prevent SROs from acting as monopolists and prioritising industry interests over public welfare (DeMarzo, Fishman and Hagerty, 2005^[137]). This model thus requires substantial administrative capacity from both public and private actors, as well as strict oversight and

integrity measures. For example, the SEC-FINRA framework enhances transparency through strict recordkeeping requirements, which prohibit FINRA members from using off-channel communication (Tami, Druehl and DeNault, 2025^[138]). OECD practices, such as cooling-off periods to address post-public employment and revolving door concerns, can also be effective in this regard (OECD, 2010^[139]).

In addition, the model gives more flexibility to industry in identifying the risks that should be addressed, with a focus on protecting citizens' wellbeing by supervising industry actions. Nevertheless, to ensure regulators can properly assess industry efforts, public actors need to have a clear understanding of the risks as well as they will form the basis of public oversight. Otherwise, the lack of clear evaluation criteria may hinder monitoring and enforcement, on which the effectiveness of the model depends.

Despite these challenges, meta self-regulation remains a cost-effective approach for public actors. It leverages private sector expertise and holds industry accountable for safe and sustainable development. Simultaneously, public oversight ensures a balance between regulatory certainty and flexibility (Walters and Wiseman, 2022^[40]). Continuous collaboration and oversight can lead to regulatory outcomes similar to those a public model would produce. However, since rules are developed within the industry, they may achieve higher levels of internalisation and compliance (Coglianese and Mendelson, 2010^[136]). Other already discussed issues related to regulatory flexibility, such as regulatory innovation, are also relevant here.

Applying the model for digital products and services

Applying meta self-regulation to digital services poses challenges. While some, for example fintech solutions, can fall within existing self-regulatory frameworks (Grennan, 2022^[140]; Walters and Wiseman, 2022^[40]), often, just as in the case of existing public regulations, they remain in a regulatory grey area with uncertain applicability of existing rules. Additionally, the creation of new, tailored SROs for emerging industries is so far limited, and can be hindered by the overlapping characteristics of many digital markets.

In such cases, public actors may intervene more directly. For example, under the Australian Online Safety Act, if no such association exists, the eSafety Commissioner can prompt its creation, with the purpose of developing a specific safety code. If no association is formed, the eSafety Commissioner has the power to commission an industry-wide standard by themselves. While not a traditional form of meta self-regulation, this illustrates how the “regulation of regulators” can take place in the circumstances created by digital markets.

In other cases, public actors can address this problem by taking some activities of SROs and constructing a legal framework within which individual company actions can be co-ordinated (see Box 3.5). Such approaches can often borrow some elements of co-regulation or other models, to achieve sufficient flexibility to allow for the logic of meta-self-regulation to be applied to less well organised markets.

Box 3.5. Regulating digital platforms' relationship with publishers in Canada and Australia

A public framework for private regulatory efforts

The Canadian Online News Act and the Australian News Media and Digital Platforms Mandatory Bargaining Code illustrate how public actors can shape the framework for meta self-regulatory efforts, even without utilising any existing SROs. Both regulations seek to manage the relationship between digital platforms and traditional media publishers by establishing a bargaining framework, intended to lessen the power imbalance and to compel agreements that achieve public objectives.

Although there are some differences, both regulations require platforms to reach agreements with publishers that clarify their relationship and the dissemination of publishers' content on the platforms. When no agreement is reached, public regulators intervene and compel both sides into arbitration. While public authorities do not mandate specific elements to be included in the agreements, they must fulfil general objectives, such as supporting the sustainability of the news industry (in Australia) or encouraging diversity in languages, ethnicities, and Indigenous communities (in Canada).

Thus, the regulation puts public actors into the position normally occupied by an SRO. It creates a framework for exchange between parties to achieve public goals and ensures equal footing among stakeholders, primarily publishers, who would otherwise face strong pressure from online platforms leveraging their dominant market position to restrict the dissemination of publishers' content.

Source: (Canada, 2023^[141]; Australia, 2021^[142]; Flew et al., 2024^[143]).

Industry self-regulation

Industry self-regulation is a model in which companies set rules and standards for the entire industry (Gunningham and Rees, 1997^[134]). This collective aspect is crucial and is often presented as a distinguishing characteristic of self-regulation itself (OECD/KDI, 2021^[10]; Black, 1996^[144]). Hence, self-organisation is also a key element of this model. Similarly to meta self-regulation, it typically involves creating an SRO and developing voluntary standards or codes of conduct.

Pure self-regulation, which entails no public intervention, remains a largely theoretical concept, akin to pure command-and-control, with public actors often more or less directly inspiring the self-regulatory effort and its results (Gunningham and Rees, 1997^[134]; Sinclair, 1997^[3]). However, under this model, private actors lead in all three regulatory domains, while public actors are still granted several important tools, such as monitoring or exerting regulatory pressure, which can significantly affect the model's effectiveness.

Table 3.6. Industry self-regulation

The role of private and public actors over regulatory processes, resulting flexibility, risks and opportunities

Stage of the regulatory process	Role		Flexibility	Risks	Opportunities
	Public actors	Private actors			
Regulatory objectives	Consulted or informed	Responsible	High	Misaligned priorities, leading to under regulation and prioritising corporate interests.	Opportunity for industry to organically fill the regulatory void in line with industry's priorities leading to higher compliance. An efficient and cost-effective way for addressing low-level risks or even medium-/high-level risks, under regulatory pressure
Regulatory means	Consulted or informed	Responsible	High	Leaving smaller companies without guidance and common goals in the case of a less well organised industry sector Risk of smaller companies facing barriers	Unilateral nature of industry-developed standards protects citizens Encouraging regulatory innovation Alignment of the regulatory means with business models leading to higher compliance
Regulatory delivery	Consulted or informed	Responsible	High	Symbolic compliance, regulatory success depending entirely on willingness of the industry Overreliance on automated and community-driven systems	Rapid development and adaptation of enforcement by the industry Regulatory standards are embedded directly in the internal processes of the companies, boosting compliance

Source: Authors' own elaboration.

The main risks to industry self-regulation arise from its voluntary character and potential conflicts of interest, as the industry simultaneously acts as rule-maker, regulator, and rule-taker. These risks mirror those outlined for meta self-regulation; however, the absence of significant formalised public influence in any regulatory domain changes their nature and impedes public intervention in cases of regulatory failure. First, self-regulatory standards must be genuinely internalised by the entire industry, otherwise problems with symbolic compliance may occur (Gunningham and Rees, 1997^[134]; Beard and Nye, 2011^[29]; OECD, 2021^[21]). This requires industry to share a common understanding of risks and perceive them as a threat to their own business models, or the industry as a whole, for example by presenting it as unsafe for consumers. This was one of the motivations for self-regulatory efforts in the nuclear sector in the US (Sinclair, 1997^[3]), and, as discussed in Chapter 4, is also valid for some digital markets. As with meta self-regulation, such commitment, as well as effective co-ordination, is more easily achieved in relatively homogeneous and small industries, where similar business models enable consensus building and make it easier to identify free-riders (King and Lenox, 2000^[30]; Cusumano, Gawer and Yoffie, 2021^[4]; Walters and Wiseman, 2022^[40]). The model was also proved to be more effective for markets in which large companies are present, as they have more compliance resources and are easier to monitor, while SMEs may struggle to comply and be more prone to free-riding, as overseeing multiple small companies can be challenging (Lynch-Wood, Williamson and Jenkins, 2009^[145]).

Second, meeting these conditions can exacerbate other risks already mentioned, such as capture by the most influential SRO members or insufficient protection for citizens. Therefore, even a limited role for public actors is crucial, as public oversight and pressure within the self-regulatory scheme are key factors in mitigating these challenges (DeMarzo, Fishman and Hagerty, 2005^[137]; Cusumano, Gawer and Yoffie, 2021^[4]).

Finally, all risks related to the construction of the SROs and industry-wide co-ordination described in meta self-regulation are also relevant for this model. Otherwise, self-regulation can lose its collective character, crucial for the effectiveness of the model and lead to the creation of multiple rules and standards, which in the digital context can take form of the “walled garden” discussed in relation to command-and-control model.

However, when an industry is motivated to self-regulate and can effectively co-ordinate, industry self-regulation can harness its expertise and understanding of the technology (Cantero Gamito and Marsden, 2024^[119]; OECD, 2024^[6]), avoid excessive costs and bureaucracy (Gunningham and Rees, 1997^[134]), and increase acceptance of and compliance with self-developed norms (Short and Toffel, 2010^[33]; Coglianese and Mendelson, 2010^[136]). Nevertheless, the voluntary nature of these schemes remains a significant limitation even when these conditions are met, often resulting in short-lived initiatives that end once companies lose interest (Gellman and Dixon, 2011^[120]).

Finally, industry self-regulation can play a sufficient role when the risks presented by a regulated service or product are low. This may be particularly relevant in the early stages of technological development, as discussed in Chapter 2, when risks are not yet fully materialised or even anticipated. Nevertheless, in such cases public actors should continue to monitor and evaluate risks to ensure the model remains aligned with their nature. For the same reasons, regulators should build expertise and capacity in parallel to self-regulatory efforts, so they are ready to intervene in case of regulatory failure.

Adapting the model for digital products and services

For more than a decade after the regulation of the digital economy began, self-regulation was the dominant approach for regulating digital markets in many jurisdictions, including the EU and the US (Koops et al., 2006^[146]). Evaluating these efforts is difficult, as there is no single “digital sector” or “industry”. Instead, multiple industries develop and apply new technologies to develop services and products, often engaging in self-regulatory efforts and co-ordinating their actions with varied success.

However, such efforts often lacked an “industry” component, lacking co-ordination and being limited to the individual regulatory activity of companies. As mentioned above, historical examples of industry self-regulation of similar services suggest that specific conditions must occur for industry-wide self-regulation to happen (see Box 3.6).

Box 3.6. The US Computer Reservation Systems self-regulation

Collective self-regulation under the public pressure

The US Computer Reservation Systems (CRSs) illustrate how political pressure can shape collective self-regulation without direct government involvement. Developed in the mid-twentieth century, it was swiftly adopted by most travel agencies. A few systems, controlled by major airlines, came to dominate the market through self-preferencing practices, such as promoting their own flights or charging other airlines for visibility in flight searches. This drew the attention of regulators and policymakers, yet the Department of Justice declined to pursue antitrust litigation, and Congress did not enact regulations. Nevertheless, under intense political pressure and amid market shifts brought about by the rise of the internet and online booking services, the industry gradually moved towards more neutral and accessible platforms, co-ordinating its actions to avoid formal public regulation.

The CRSs case exemplifies self-regulation in a setting akin to modern digital platforms. This shift took place under specific conditions, including multiple investigations, unsuccessful legislative action, various lawsuits, and the broader transition to individual online booking. Moreover, a small number of similar companies controlled the sector, enabling them to align objectives and co-ordinate effectively. Ultimately, opening their platforms proved more profitable, as relinquishing some oligopolistic control expanded their user base.

Given the growing popularity of the blended models described in the paper, and limited use of models such as command-and-control, which lower the regulatory pressure, it is possible that many digital services providers today may prefer to participate in blended models, giving them a relative degree of flexibility over engaging in self-organisation of the whole sector.

Source: (Cusumano, Gawer and Yoffie, 2021^[4]).

One clear example of a sector organised in a more traditional way is digital advertising, with organisations in the EU (e.g., European Interactive Digital Advertising Alliance) and the US providing examples (e.g., Digital Advertising Alliance) (EDAA, 2025^[147]; DAA, 2025^[148]). By contrast, social media platforms lack strong SROs, with existing organisations serving different roles, focusing largely on sharing data and expertise in specific areas, such as combating child sexual abuse materials (CSAM) or terrorist content, for example with initiatives like Global Internet Forum for Counter Terrorism (GIFCT, 2025^[149]). These industry efforts to co-ordinate and share data are useful to support individual regulatory actions, rather than creating sector-wide frameworks. In other areas, frameworks are still largely co-developed together with public actors, as seen with the EU Strengthened Code of Conduct on Disinformation analysed above.

Industry self-regulation can be also problematic for some digital services providers. For example, while in some markets, such as social media platforms, large, homogenous companies are present, it is not the case for multiple other sectors of digital economy, for example small AI start-ups exploring innovative medical services. This limits the possibility of self-organisation and sufficient monitoring as mentioned above.

In the digital context, industry self-regulation can also create a risk of private actors' overreliance on automation (Durach, Bârgăoanu and Nastasiu, 2020^[150]; Llansó, 2020^[151]) or shifting compliance burdens to users (Selin, 2016^[152]). This is well illustrated by the case of content moderation, where both AI systems and user-driven moderation are often employed, providing a cheaper alternative to more accurate human moderation (Levi et al., 2025^[153]).

4 Adapting the spectrum of regulatory models to the digital economy

Regulatory governance addresses issues too complex and context-dependent for simple, universal solutions. As shown in this report, each model of regulatory governance involves specific trade-offs and investments, while existing within a particular environment, parts of which, such as industry or market characteristics, lie outside the control of policymakers and regulators.

The purpose of this work is not to identify an ideal regulatory model for the digital economy, but to guide more informed decisions regarding the regulatory governance strategy in digital contexts. As noted across the report, digital services have certain characteristics that make identification of the most suitable regulatory models a challenge. Chapter 2 therefore sets out general criteria that can help policymakers and regulators choose an appropriate model, illustrated through examples from the digital economy. Chapter 3 then shows how different models, or particular elements of them, can be adapted to the specific features of this area.

Taken together, the analysis presented in these two chapters makes it possible to draw a more integrated picture of the regulatory governance decision-making process. By combining the contextual assessment considerations developed in Chapter 2 with the model characteristics and trade-offs described in Chapter 3, policymakers can move beyond identifying relevant assessment criteria in isolation and instead trace how the outcomes of those assessments – regarding technological maturity, market structure, private actor capacity, and public institutional readiness – can systematically point toward more public or more private models of regulatory governance. Table 4.1 brings these two dimensions together, offering a consolidated reference that links contextual findings to governance model choices, and which can support more transparent and structured regulatory decision-making across a range of sectors and policy contexts.

That said, by offering a general methodology and a set of model characteristics key elements of this paper can be tailored to analyse other areas and markets beyond the digital economy. Common best practices and principles remain crucial for choosing and implementing tailored solutions in any sector, coupled with unique considerations and assessments based on their context. The OECD Recommendations on agile regulation (OECD, 2024^[96]) and regulatory policy and governance (OECD, 2012^[25]) can help policymakers and regulators apply the tools and practices outlined in Chapter 4, enabling them to draw on the advantages of different models while limiting the risks these may entail. Specific instruments such as the Regulatory Sandbox Toolkit (OECD, 2025^[154]), can further allow them to ensure the regulatory framework they create is adapting to the changing environment described in this report.

Table 4.1. Key considerations for choosing a regulatory model

Potential considerations when choosing a model	Regulatory model	
	More public-led	More private-led
Impact on innovation		
Stage of technological development of the sector	More appropriate with increasing product maturity	More appropriate at the beginning of product's development and application, when risks are not yet assessed.
Risk posed by innovation	More justified for risks categorised as "high"	More justified for risks categorised as "low"
Nature of private actors and markets		
Private actor willingness and capacity for regulatory governance	More justified when industry lacks incentives for action or capacity in past regulatory efforts	Requires the presence of general incentives motivating private regulatory action, proven capacities to meet the specific requirements of a given private regulatory model, as well as a satisfactory level of trust between private and public actors.
Market characteristics	More justified for less homogenous and organised markets, with multiple, especially small, companies	More justified for heterogenous, well-organised markets, with small number of big companies
Integrity, transparency and accountability	Requires special focus on maintaining integrity in lobbying and public consultations	Requires special focus on ensuring equal access to the process of creating and managing private or collaborative regulatory system to prevent regulatory capture
Public institutional readiness		
Expertise and skills	Requires more sophisticated, in-depth knowledge of the regulated area for effective creation and enforcement of given regulatory models	Requires fundamental expertise and knowledge from public actors to assess risk and monitor private actors' efforts.
Digital tools	Higher sophistication of digital tools required, particularly to deliver and enforce regulations via tools that keep up with innovative products or services being regulated	A variety of digital tools still required from public actors for effective collection of data and monitoring of private regulatory efforts
Institutional co-ordination	Clear definitions of mandates, as well as mechanisms for joined-up governance and co-ordinated enforcement strategies	Monitoring and oversight responsibilities should be implemented together with clear allocation of roles of the private and public actors

Source: Authors' own elaboration.

Key takeaways for the digital economy

A regulatory governance perspective on specific cases and developments in the digital economy suggests three broad trends. These findings should be treated as preliminary and may vary across political contexts. Even so, they are closely linked to the nature of digital products and services and the regulatory challenges they create. They may therefore still help policymakers and regulators better understand the options available to them and the investments these may require.

A false dichotomy between command-and-control and industry self-regulation

While the spectrum presented in this paper confirms that viewing regulation through a simple dichotomy between command-and-control and industry self-regulation is overly simplistic in general (Sinclair, 1997^[3]), the analysis in Chapter 2 and Chapter 3 suggests that, the most private-led and the most public-led models of regulatory governance are the least likely to be used for digital services regulation.

In practice, the nature of many parts of the digital economy means that regulatory governance often needs to strike a balance between different features of digital services. Policymakers may need to weigh the effects of regulation on innovation-driven products against the significant risks their novelty can introduce, for example through new business models or insufficient consumer protection. The fast-changing nature

of emerging digital technologies means that regulators may be aiming at a moving target, creating a need for flexible and adaptable regulatory approaches. In some areas, they deal with large, though not necessarily organised for self-regulation, companies, in others, with small start-ups that lack extensive policy expertise. The regulation must be also designed and implemented by entities with sufficient skills and understanding of services that often rely on highly sophisticated technology.

These factors make both command-and-control and industry self-regulation particularly difficult to apply in the digital economy. On the one hand, a purely public approach requires substantial technical capability, not only to design regulation properly but also to deliver and enforce it. The latter is especially challenging for services such as social media, for example, which relies on complex algorithms and generate enormous volumes of data, creating a substantial asymmetry of information between regulators and service providers.

On the other hand, industry self-regulation is difficult because digital services are heterogeneous and often cross-sectoral, which can make it hard to identify a distinct “industry” able to organise itself. While some areas of the digital economy can be defined, they are rarely sufficiently self-organised, as in the case of social media platforms, or they include too many small firms. There are some exceptions, such as fintech services, which in some countries may fall under self-regulatory organisation schemes. However, this is less common for the most innovative or disruptive services, which are often not easily integrated into existing self-regulatory frameworks. More broadly, the prevalence of blended models may also reduce pressure for industry-level self-regulation, as many firms may prefer co-regulation or similar approaches that offer more direct engagement with public actors, a high degree of flexibility, and public support, for example in co-ordination.

For these reasons, as Chapter 3 illustrates, it is difficult to identify clear cases in which either of these two models is applied to digital services. Instead, policymakers often adapt more established or more tailored blended models, which may be better suited to the nature of digital services and the digital economy more broadly.

Delegation and oversight in blended regulatory governance

When examining how policymakers use blended models to regulate digital services, it appears that public actors involve private actors mainly in regulatory means and delivery, while retaining responsibility for regulatory objectives. At the same time, they continue to oversee, and often manage, selected processes across the regulatory cycle. This is often done by co-ordinating private regulatory initiatives, stepping into the role often played in different industries by an industry’s SRO.

Public responsibility for setting regulatory objectives is common across all analysed models, except industry self-regulation, which stems also from public actors effectively holding a monopoly on defining broader policy goals. However, the way in which the goals are formulated appears to reflect the nature of digital services and some broader structural features of the digital economy. These are often framed in general terms and need to be operationalised by regulators or by private actors involved in regulatory means. For example, under more performance-based models, it may fall to these actors to define what counts as “necessary measures”. This more general formulation of objectives may reflect an effort to create technology-neutral frameworks that remain flexible as innovation evolves. It may also stem from the wider challenge of defining clear measurable outputs for digital services.

It is at the level of regulatory means and delivery that private actors become most heavily involved, as the technical expertise required for defining regulatory compliance processes often lies with industry. However, this delegation is rarely unconditional, with public actors tending to move into an oversight role. They retain influence over how objectives are translated into practice, setting a general framework and a baseline for private action, supported through guidance documents and self-assessment tools (as under the Australian Online Safety Act or the GDPR), and co-ordinating private efforts in a way that resembles that of a sector

SRO. In similar way, public actors can create and organise the market for a given service or its underlying technology, as in the case of 5G auctions or the market-based solutions in the UK and US analysed in Chapter 3. In the case of regulatory delivery, where information asymmetries make traditional inspections difficult, public actors can monitor companies' self-policing through transparency reporting and risk assessments. They can also maintain ongoing contact with firms, as with the Australian eSafety Commissioner, or embed designated functions within companies themselves, as with data protection officers under the GDPR.

Taken together, these patterns suggest a coherent public-sector response to the structural features of digital markets, based on delegating where technical complexity or information asymmetry makes direct action impractical, but retaining the levers needed to set direction, ensure consistency, and hold private actors accountable. When implemented well, such models can draw on the flexibility and expertise of private-led regulatory processes while preserving the safeguards that public oversight provides.

What blended regulatory models mean for regulators

The trends identified above carry direct implications for how regulators approach the digital economy and, more specifically, for what is expected of them. The choice of a regulatory model is not merely a policy preference; it implicitly assigns a particular role to the regulator and generates corresponding institutional and capacity requirements. These requirements deserve explicit recognition and deliberate investment.

Where models situated in the middle of the governance spectrum are applied to digital services, two sets of institutional demands stand out. First, given the importance of regulatory objectives across these models, both regulators and policymakers should consider strengthening mechanisms that help them better understand the services being regulated. This points to the value of strategic intelligence functions, regulatory experimentation, and related instruments discussed in Chapter 2. Second, the regulator's evolving role in designing regulatory means and in delivery should be also acknowledged. Where responsibility for these functions is delegated to private actors, regulators must be equipped to act as effective overseers of the wider process, co-ordinating industry efforts, issuing targeted guidance for smaller firms and start-ups, and developing new approaches to supervision that draw on data and reporting provided by private actors. Throughout, appropriate safeguards will be needed to ensure that public-private collaboration remains transparent and accountable.

It should be noted, however, that these observations reflect a selection of trends associated with current market conditions and the particular characteristics of the digital economy. The increasing prevalence of blended models does not imply that they represent a permanent or universal solution. As a foundational assumption of this report, regulatory approaches must be capable of evolving alongside the markets they govern. This applies in both directions. Sustained regulatory pressure, combined with clear signals that industry self-regulation is a preferred model, could make certain parts of the digital economy more amenable to collective self-regulatory arrangements. Equally, command-and-control approaches remain available in principle, though their effective application to digital services would require not only a detailed understanding of relevant technologies and their development trajectories, but also substantial public capacity to design and implement precise standards.

The way forward

Translating these insights into practice requires deliberate and sustained investment by regulators across several interconnected areas, included in the OECD Recommendation on Agile Regulatory Governance to Harness Innovation.

- First, strategic intelligence functions, including horizon scanning, foresight and regulatory observatories, enable regulators to keep pace with fast-evolving markets and to anticipate, rather than merely react to, emerging risks.
- Second, structured sector dialogue and trust-building with private actors are essential for blended models to function effectively, particularly where technical expertise and compliance-relevant data lie predominantly in private hands. This requires both formal mechanisms of engagement and safeguards to prevent capture, as outlined in Chapter 2.
- Third, regulatory experimentation, through sandboxes, policy prototyping and testbeds, allows regulators to test governance approaches alongside innovators and to refine their choice of model as evidence accumulates.
- Fourth, investment in digital tools and in-house technical skills within regulators themselves is indispensable: without sufficient internal capacity, regulators cannot credibly evaluate industry self-assessments, scrutinise algorithmic systems, or co-ordinate supervision across jurisdictions and sectors. These investments are not alternatives to selecting a regulatory governance model, they are the foundations that determine whether any chosen model can deliver its intended outcomes, and that allow the assessment process to move from a one-off exercise into an ongoing regulatory practice.

Building on this, further research could deepen the analysis in several ways. Comparative empirical work on how specific blended models perform against their intended objectives over time would help refine the trade-offs identified here, particularly as frameworks governing emerging technologies such as generative AI mature and are tested in implementation. Closer examination of the institutional capacities and oversight arrangements leading to successful applications of blended models would also be valuable, as would further work on how transparency and accountability mechanisms can be adapted to the distinctive information asymmetries of digital markets.

Ultimately, the spectrum of regulatory models presented in this paper is not a presentation of fixed choices but a framework for navigating an evolving landscape. The task for policymakers and regulators is less about identifying a single right model than about building the institutional conditions to choose well, adjust when circumstances change, and sustain the legitimacy of regulatory action as technologies, markets and societal expectations continue to evolve.

References

- ACCC (2025), *Digital platform services inquiry 2020-25*, <https://www.accc.gov.au/inquiries-and-consultations/digital-platform-services-inquiry-2020-25/march-2025-final-report>. [36]
- Albareda, L. (2008), “Corporate responsibility, governance and accountability: From self-regulation to co-regulation”, *Corporate Governance*, Vol. 8/4, <https://doi.org/10.1108/14720700810899176>. [26]
- ARC2 (2025), “OLIVIA. Virtual assistant for GDPR compliance”, <https://olivia-gdpr-arc.eu/en>. [48]
- Askim, J. et al. (2011), “One-stop shops for social welfare: The adaptation of an organizational form in three countries”, *Public Administration*, Vol. 89/4, <https://doi.org/10.1111/j.1467-9299.2011.01933.x>. [52]
- Atsu, F. and S. Adams (2023), “Financial development and innovation: Do institutions and human capital matter?”, *Heliyon*, Vol. 9/8, p. e19015, <https://doi.org/10.1016/j.heliyon.2023.e19015>. [18]
- Australia (2025), *Age Assurance Technology Trial— Final Report*, <https://www.infrastructure.gov.au/departments/media/publications/age-assurance-technology-trial-final-report>. [94]
- Australia (2025), *Online Safety (Age-Restricted Social Media Platforms) Rules 2025*, <https://www.legislation.gov.au/F2025L00889/latest/text>. [92]
- Australia (2024), *Online Safety Amendment (Social Media Minimum Age) Act 2024*, <https://www.legislation.gov.au/C2024A00127/asmade/text>. [91]
- Australia (2023), *Schedule 2 – App Distribution Services Online Safety Code (Class 1A and Class 1B Material)*, <https://www.esafety.gov.au/sites/default/files/2023-06/Schedule-2%E2%80%93App-Distribution-Services-Online-Safety-Code-%28Class-1A-and-Class-1B%20Material%29.pdf?v=1741626994564>. [55]
- Australia (2021), *Online Safety Act No. 76, 2021*. [123]
- Australia (2021), *Treasury Laws Amendment (News Media and Digital Platforms Mandatory Bargaining Code) Act 2021*. [142]
- Balleisen, E. and M. Eisner (2009), “The Promise and Pitfalls of Co-Regulation: How Governments Can Draw on Private Governance for Public Purpose”, *New Perspectives on Regulation*. [88]

- Beard, F. and C. Nye (2011), “A History of the Media Industry’s Self-Regulation of Comparative Advertising”, *Journalism History*, Vol. 37/2, <https://doi.org/10.1080/00947679.2011.12062850>. [29]
- Benoliel, D. (2004), *Technological standards, Inc.: Rethinking cyberspace regulatory epistemology*, <https://doi.org/10.2307/3481318>. [62]
- Benoliel, D. (2004), *Technological standards, Inc.: Rethinking cyberspace regulatory epistemology*, <https://doi.org/10.2307/3481318>. [131]
- Berger, M., A. Dechezleprêtre and M. Fadic (2024), “What is the role of Government Venture Capital for innovation-driven entrepreneurship?”, *OECD Science, Technology and Industry Working Papers*, No. 2024/10, OECD Publishing, Paris, <https://doi.org/10.1787/6430069e-en>. [106]
- Black, J. (2001), “Decentring Regulation: Understanding the Role of Regulation and Self-Regulation in a ‘Post-Regulatory’ World”, *Current Legal Problems*, Vol. 54/1, <https://doi.org/10.1093/clp/54.1.103>. [2]
- Black, J. (1996), “Constitutionalising Self-Regulation”, *The Modern Law Review*, Vol. 59/1, <https://doi.org/10.1111/j.1468-2230.1996.tb02064.x>. [144]
- Black, J., M. Hopper and C. Band (2007), “Making a success of Principles-based regulation”, *Law and Financial Markets Review*, Vol. 1/3, <https://doi.org/10.1080/17521440.2007.11427879>. [86]
- Borz, G. et al. (2024), “The EU soft regulation of digital campaigning: regulatory effectiveness through platform compliance to the code of practice on disinformation”, *Policy Studies*, Vol. 45/5, <https://doi.org/10.1080/01442872.2024.2302448>. [126]
- Bradford, A. (2024), “The False Choice Between Digital Regulation and Innovation”, *SSRN Electronic Journal*, <https://doi.org/10.2139/ssrn.4753107>. [16]
- Bradford, A. (2020), *The Brussels Effect*, Oxford University Press New York, <https://doi.org/10.1093/oso/9780190088583.001.0001>. [82]
- Cabanillas, C., M. Resinas and A. Ruiz-Cortés (2012), “Automated Resource Assignment in BPMN Models Using RACI Matrices”, in *Lecture Notes in Computer Science, On the Move to Meaningful Internet Systems: OTM 2012*, Springer Berlin Heidelberg, Berlin, Heidelberg, https://doi.org/10.1007/978-3-642-33606-5_5. [59]
- Canada (2023), *Online News Act*, S.C. 2023, c. 23. [141]
- Cantero Gamito, M. and C. Marsden (2024), “Artificial intelligence co-regulation? The role of standards in the EU AI Act”, *International Journal of Law and Information Technology*, Vol. 32/1, <https://doi.org/10.1093/ijlit/eaee011>. [119]
- Cantero Gamito, M. and C. Marsden (2024), “Artificial intelligence co-regulation? The role of standards in the EU AI Act”, *International Journal of Law and Information Technology*, Vol. 32/1, <https://doi.org/10.1093/ijlit/eaee011>. [132]
- Chee, F. (2024), “Apple to delay launch of AI-powered features in Europe, blames EU tech rules”, *Reuters*. [84]
- Coalition for Content Provenance and Authenticity (2025), *C2PA*, <https://c2pa.org/>. [90]

- Coglianese, C. (2016), "Performance-based regulation: concepts and challenges", in *Comparative Law and Regulation*, Edward Elgar Publishing, <https://doi.org/10.4337/9781782545613.00028>. [64]
- Coglianese, C. and E. Mendelson (2010), "Meta-Regulation and Self-Regulation", in *The Oxford Handbook of Regulation*, <https://doi.org/10.1093/oxfordhb/9780199560219.003.0008>. [136]
- Coglianese, C. and J. Nash (2017), *The Law of the Test: Performance-Based Regulation and Diesel Emissions Control*, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2888990. [9]
- Cole, D. and P. Grossman (2018), "When is command-and-control efficient? Institutions, technology, and the comparative efficiency of alternative regulatory regimes for environmental protection", in *The Theory and Practice of Command and Control in Environmental Policy*, <https://doi.org/10.4324/9781315197296-7>. [80]
- Cuerva, M., Á. Triguero-Cano and D. Córcoles (2014), "Drivers of green and non-green innovation: empirical evidence in Low-Tech SMEs", *Journal of Cleaner Production*, Vol. 68, pp. 104-113, <https://doi.org/10.1016/j.jclepro.2013.10.049>. [115]
- Cusumano, M., A. Gawer and D. Yoffie (2021), "Can Self-Regulation Save Digital Platforms?", *SSRN Electronic Journal*, <https://doi.org/10.2139/ssrn.3900137>. [4]
- DAA (2025), "DAA website", <https://digitaladvertisingalliance.org/>. [148]
- Davies, P. (2024), "OpenAI releases AI video creator Sora but it won't be coming to Europe yet", *Europnews*, <https://www.euronews.com/next/2024/12/10/openai-releases-ai-video-creator-sora-but-it-wont-be-coming-to-europe-yet> (accessed on 17 June 2025). [85]
- DeMarzo, P., M. Fishman and K. Hagerty (2005), "Self-regulation and government oversight", *Review of Economic Studies*, Vol. 72/3, <https://doi.org/10.1111/j.1467-937X.2005.00348.x>. [137]
- Drake, W., V. Cerf and W. Kleinwachter (2016), "Internet Fragmentation: An Overview", *Future of the Internet Initiative White Paper* January. [65]
- DRCF (2025), *Digital Regulation Cooperation Forum - Home*, <https://www.drcf.org.uk/>. [50]
- Durach, F., A. Bârgăoanu and C. Nastasiu (2020), "Tackling disinformation: EU regulation of the digital space", *Romanian Journal of European Affairs*, Vol. 20/1. [150]
- EDAA (2025), "EDAA website", <https://edaa.eu/>. [147]
- E-Estonia (2025), "X-Road", <https://e-estonia.com/solutions/x-road-interoperability-services/x-road/>. [49]
- eSafety Commissioner (2025), "Industry codes and standards", <https://www.esafety.gov.au/industry/codes>. [125]
- eSafety Commissioner (2025), "Register of industry codes and industry standards for online safety", <https://www.esafety.gov.au/industry/codes/register-online-industry-codes-standards>. [124]
- eSafety Commissioner (n.d.), *How to assess if a service is an age-restricted social media platform*, <https://www.esafety.gov.au/about-us/industry-regulation/social-media-age-restrictions/assessment>. [93]

- eSafety Commissioner (n.d.), *Which platforms are age-restricted?*, [95]
<https://www.esafety.gov.au/about-us/industry-regulation/social-media-age-restrictions/which-platforms-are-age-restricted>.
- Escobar, J. et al. (2023), "The 5G spectrum auction in Chile", *Telecommunications Policy*, [108]
 Vol. 47/7, p. 102580, <https://doi.org/10.1016/j.telpol.2023.102580>.
- EU (2026), *Codes of conduct under the Digital Services Act*, <https://digital-strategy.ec.europa.eu/en/policies/dsa-codes-conduct>. [128]
- EU (2025), *The Code of conduct on countering illegal hate speech online* +, <https://digital-strategy.ec.europa.eu/en/library/code-conduct-countering-illegal-hate-speech-online>. [129]
- EU (2025), *The Code of Conduct on Disinformation*, <https://digital-strategy.ec.europa.eu/en/library/code-conduct-disinformation>. [130]
- EU (2024), "Regulation 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 an", <https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>. [67]
- EU (2022), *Directive (EU) 2022/2380 of the European Parliament and of the Council of 23 November 2022 amending Directive 2014/53/EU on the harmonisation of the laws of the Member States relating to the making available on the market of radio equipment*, <https://eur-lex.europa.eu/eli/dir/2022/2380/oj/eng>. [56]
- EU (2022), *Regulation 2022/2065 of the European Parliament and of the Council of 19 October 2022 on a Single Market For Digital Services and amending Directive 2000/31/EC (Digital Services Act)*, <https://eur-lex.europa.eu/eli/reg/2022/2065/oj/eng>. [58]
- EU (2021), *The assembly of the signatories to strengthen the Code of Practice on disinformation*, <https://digital-strategy.ec.europa.eu/en/news/assembly-signatories-strengthen-code-practice-disinformation>. [122]
- EU (2017), "Guidelines on Data Protection Officers ('DPOs')", [35]
<https://ec.europa.eu/newsroom/article29/items/612048>.
- EU (2016), *Regulation 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation)*, <https://eur-lex.europa.eu/eli/reg/2016/679/oj/eng>. [34]
- EU (n.d.), *The 2022 Code of Practice on Disinformation*, <https://digital-strategy.ec.europa.eu/en/policies/code-practice-disinformation>. [121]
- European Commission (2023), *Better regulation toolbox*, European Commission, https://commission.europa.eu/law/law-making-process/planning-and-proposing-law/better-regulation/better-regulation-guidelines-and-toolbox_en (accessed on 14 January 2025). [72]
- European Commission: Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs, Economisti Associati, Fraunhofer, IPSOS, Trinomics and Vazquez, Y. (2021), *Impact assessment study to assess unbundling of chargers – Final report*, <https://doi.org/doi/10.2873/788086>. [79]

- European Union (2023), *Commission Delegated Regulation (EU) 2023/1717 of 27 June 2023 amending Directive 2014/53/EU of the European Parliament and of the Council as regards the technical specifications for the charging receptacle and charging communication protocol for all the*, https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=uriserv%3AOJ.L_.2023.223.01.0001.01.ENG. [76]
- European Union (2022), *Directive (EU) 2022/2380 of the European Parliament and of the Council of 23 November 2022 amending Directive 2014/53/EU on the harmonisation of the laws of the Member States relating to the making available on the market of radio equipment*, https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=uriserv%3AOJ.L_.2022.315.01.0030.01.ENG&toc=OJ%3AL%3A2022%3A315%3ATOC. [77]
- European Union (2019), *Commission Regulation (EU) 2019/1782 of 1 October 2019 laying down ecodesign requirements for external power supplies pursuant to Directive 2009/125/EC of the European Parliament and of the Council and repealing Commission Regulation (EC) No 278/2009*, <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A32019R1782>. [78]
- Fabrizio, K. (2012), “The Effect of Regulatory Uncertainty on Investment: Evidence from Renewable Energy Generation”, *Journal of Law, Economics, and Organization*, Vol. 29/4, pp. 765-798, <https://doi.org/10.1093/jleo/ews007>. [19]
- Finck, M. (2018), “Blockchains: Regulating the Unknown”, *German Law Journal*, Vol. 19/4, pp. 665-692, <https://doi.org/10.1017/S2071832200022847>. [13]
- Fisher, E. (2010), “Risk Regulatory Concepts and the Law”, in *Risk and Regulatory Policy: Improving the Governance of Risk*, OECD Publishing, Paris, <https://doi.org/10.1787/9789264082939-6-en>. [103]
- Flew, T. et al. (2024), “Digital platforms and the future of news: regulating publisher-platform relations in Australia and Canada”, *Information, Communication & Society*, Vol. 27/16, pp. 2727-2743, <https://doi.org/10.1080/1369118X.2023.2291462>. [143]
- Gellman, R. and P. Dixon (2011), *Many Failures: A Brief History of Privacy Self-Regulation*, WPF, <https://www.worldprivacyforum.org/2011/10/report-many-failures-a-brief-history-of-privacy-self-regulation/> (accessed on 14 October 2024). [120]
- GIFCT (2025), “GIFCT website”, <https://gifct.org/>. [149]
- Gornet, M. and W. Maxwell (2024), “The European approach to regulating AI through technical standards”, *Internet Policy Review*, Vol. 13/3, <https://doi.org/10.14763/2024.3.1784>. [133]
- Grennan, J. (2022), “FinTech Regulation in the United States: Past, Present, and Future”, *SSRN Electronic Journal*, <https://doi.org/10.2139/ssrn.4045057>. [140]
- Griffin, R. and C. Vander Maelen (2023), “Codes of Conduct in the Digital Services Act: Exploring the Opportunities and Challenges”, *SSRN Electronic Journal*, <https://doi.org/10.2139/ssrn.4463874>. [75]
- Gunningham, N. and J. Rees (1997), “Industry self-regulation: An institutional perspective”, *Law and Policy*, Vol. 19/4, <https://doi.org/10.1111/1467-9930.t01-1-00033>. [134]
- Hirsch, D. (2010), “The Law and Policy of Online Privacy : Regulation, Self-regulation or Co-Regulation?”, *Seattle University Law Review*, Vol. 34/439. [117]

- Hodges, C. (2016), “Ethical Business Regulation: Understanding the Evidence”, *BRDO Better Regulation Delivery Office* February. [60]
- Howcroft, E. (2025), “European securities regulator warns about crypto firms misleading customers”, *Reuters*, <https://www.reuters.com/sustainability/boards-policy-regulation/european-securities-regulator-warns-about-crypto-firms-misleading-customers-2025-07-11> (accessed on 11 July 2025). [31]
- Jenik, I. and K. Lauer (2017), “Regulatory Sandboxes and Financial Inclusion”, *CGAP Working Paper* October. [14]
- Jové-Llopis, E. and A. Segarra-Blasco (2018), “Eco-innovation strategies: A panel data analysis of Spanish manufacturing firms”, *Business Strategy and the Environment*, Vol. 27/8, pp. 1209-1220, <https://doi.org/10.1002/bse.2063>. [116]
- Kamara, I. (2017), “Co-regulation in EU personal data protection: the case of technical standards and the privacy by design standardisation ‘mandate’”, *European Journal of Law and Technology*, Vol. 8/1, <https://www.ejlt.org/index.php/ejlt/article/view/545> (accessed on 14 January 2025). [61]
- King, A. and M. Lenox (2000), “Industry self-regulation without sanctions: The chemical industry’s responsible care program”, *Academy of Management Journal*, Vol. 43/4, <https://doi.org/10.2307/1556362>. [30]
- Komaitis, K. (2023), *Internet Fragmentation: Why It Matters for Europe*. [83]
- Koops, B. et al. (2006), “Should Self-Regulation Be The Starting Point?”, https://doi.org/10.1007/978-90-6704-665-7_5. [146]
- Kramcsák, P. (2023), “Can legitimate interest be an appropriate lawful basis for processing Artificial Intelligence training datasets?”, *Computer Law & Security Review*, Vol. 48, p. 105765, <https://doi.org/10.1016/j.clsr.2022.105765>. [98]
- Levi, A. et al. (2025), “AI vs. Human Moderators: A Comparative Evaluation of Multimodal LLMs in Content Moderation for Brand Safety”, *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 5965-5973, <https://doi.org/10.48550/arXiv.2508.05527>. [153]
- Levin, J. and A. Skrzypacz (2016), *Properties of the Combinatorial Clock Auction*, <https://doi.org/10.1257/aer.20141212>. [109]
- Li, W. et al. (2018), “The effect of command-and-control regulation on environmental technological innovation in China: a spatial econometric approach”, *Environmental Science and Pollution Research*, <https://doi.org/10.1007/s11356-018-3678-3>. [71]
- Llansó, E. (2020), *No amount of “AI” in content moderation will solve filtering’s prior-restraint problem*, <https://doi.org/10.1177/2053951720920686>. [151]
- Lynch-Wood, G., D. Williamson and W. Jenkins (2009), “The over-reliance on self-regulation in CSR policy”, *Business Ethics: A European Review*, Vol. 18/1, <https://doi.org/10.1111/j.1467-8608.2009.01548.x>. [145]
- Macedoni, L. and A. Weinberger (2024), “Lobbying for Regulations: When Big Business Says Yes”, https://cmepg.gmu.edu/wp-content/uploads/2024/10/lobbying_v2.pdf (accessed on 11 July 2025). [32]

- Malloy, T. (2010), *The social construction of regulation: Lessons from the war against command and control*. [68]
- Marchant, G., L. Tournas and C. Gutierrez (2020), "Governing Emerging Technologies Through Soft Law: Lessons for Artificial Intelligence", *Jurimetrics*, Vol. 61/1. [127]
- Mazaheri, M. et al. (2022), "Market-based instruments and sustainable innovation: A systematic literature review and critique", *Journal of Cleaner Production*, Vol. 373, p. 133947, <https://doi.org/10.1016/J.JCLEPRO.2022.133947>. [102]
- McAllister, L. (2014), "Harnessing Private Regulation", *Michigan Journal of Environmental & Administrative Law* 3.2, <https://doi.org/10.36640/mjeal.3.2.harnessing>. [74]
- Microsoft (2018), *Facial recognition technology: The need for public regulation and corporate responsibility*, <https://blogs.microsoft.com/on-the-issues/2018/07/13/facial-recognition-technology-the-need-for-public-regulation-and-corporate-responsibility/> (accessed on 21 July 2025). [23]
- Milgrom, P. (2004), *Putting auction theory to work*, <https://doi.org/10.1017/CBO9780511813825>. [110]
- Miracle, G. and T. Nevett (1988), "A Comparative History of Advertising Self-regulation in the UK and the USA", *European Journal of Marketing*, Vol. 22/4, pp. 7-23, <https://doi.org/10.1108/EUM0000000005278>. [39]
- Newman, A. and D. Bach (2004), *Self-regulatory trajectories in the shadow of public power: Resolving digital dilemmas in Europe and the United States*, <https://doi.org/10.1111/j.0952-1895.2004.00251.x>. [38]
- OECD (2025), "Agile mechanisms for responsible technology development", *OECD Science, Technology and Industry Policy Papers*, No. 176, OECD Publishing, Paris, <https://doi.org/10.1787/2a35358e-en>. [15]
- OECD (2025), *OECD Regulatory Policy Outlook 2025*, OECD Publishing, Paris, <https://doi.org/10.1787/56b60e39-en>. [43]
- OECD (2025), *Regulatory sandbox toolkit: A comprehensive guide for regulators to establish and manage regulatory sandboxes effectively*, OECD Publishing, Paris, <https://doi.org/10.1787/de36fa62-en>. [154]
- OECD (2025), *Reinforcing Regulatory Frameworks through Standards, Measurements and Assurance: Making Better Use of Quality Infrastructure in Policymaking*, OECD Publishing, Paris, <https://doi.org/10.1787/f398be90-en>. [54]
- OECD (2025), *STIP Compass*, <https://stip.oecd.org/stip/interactive-dashboards/policy-initiatives/2023%2Fdata%2FpolicyInitiatives%2F26890> (accessed on 21 July 2025). [155]
- OECD (2024), *Anti-Corruption and Integrity Outlook 2024*, OECD Publishing, Paris, <https://doi.org/10.1787/968587cd-en> (accessed on 9 July 2025). [46]
- OECD (2024), *Facts not Fakes: Tackling Disinformation, Strengthening Information Integrity*, OECD Publishing, Paris, <https://doi.org/10.1787/d909ff7a-en>. [6]

- OECD (2024), "Framework for Anticipatory Governance of Emerging Technologies", *OECD Science, Technology and Industry Policy Papers*, No. 165, OECD Publishing, Paris, <https://doi.org/10.1787/0248ead5-en>. [24]
- OECD (2024), *Recommendation of the Council for Agile Regulatory Governance to Harness Innovation*, <http://OECD/LEGAL/0464>. [96]
- OECD (2024), *Recommendation of the Council on Transparency and Integrity in Lobbying and Influence*, <http://OECD/LEGAL/0379>. [44]
- OECD (2024), "Towards digital safety by design for children", *OECD Digital Economy Papers*, No. 363, OECD Publishing, Paris, <https://doi.org/10.1787/c167b650-en>. [7]
- OECD (2024), "Transparency reporting on terrorist and violent extremist content online: Fourth edition", *OECD Digital Economy Papers*, No. 367, OECD Publishing, Paris, <https://doi.org/10.1787/901cb8cf-en>. [8]
- OECD (2023), "The Impact of R&D tax incentives: Results from the OECD microBeRD+ project", *OECD Science, Technology and Industry Policy Papers*, No. 159, OECD Publishing, Paris, <https://doi.org/10.1787/1937ac6b-en>. [105]
- OECD (2022), *Equipping Agile and Autonomous Regulators*, The Governance of Regulators, OECD Publishing, Paris, <https://doi.org/10.1787/7dcb34c8-en>. [81]
- OECD (2021), *Lobbying in the 21st Century: Transparency, Integrity and Access*, OECD Publishing, Paris, <https://doi.org/10.1787/c6d8eff8-en>. (accessed on 9 July 2025). [47]
- OECD (2021), *OECD Regulatory Policy Outlook 2021*, OECD Publishing, Paris, <https://doi.org/10.1787/38b0fdb1-en>. [21]
- OECD (2021), "Recommendation of the Council for Agile Regulatory Governance to Harness Innovation", <https://legalinstruments.oecd.org/en/instruments/OECD-LEGAL-0464>. [5]
- OECD (2020), *One-Stop Shops for Citizens and Business*, OECD Best Practice Principles for Regulatory Policy, OECD Publishing, Paris, <https://doi.org/10.1787/b0b0924e-en>. [20]
- OECD (2020), *Regulatory Impact Assessment*, OECD Best Practice Principles for Regulatory Policy, OECD Publishing, Paris, <https://doi.org/10.1787/7a9638cb-en>. [27]
- OECD (2018), *OECD Regulatory Enforcement and Inspections Toolkit*, OECD Publishing, Paris, <https://doi.org/10.1787/9789264303959-en>. [57]
- OECD (2017), *Preventing Policy Capture: Integrity in Public Decision Making*, OECD Public Governance Reviews, OECD Publishing, Paris, <https://doi.org/10.1787/9789264065239-en>. [42]
- OECD (2015), "Industry Self Regulation: Role and Use in Supporting Consumer Interests", *OECD Digital Economy Papers*, No. 247, OECD Publishing, Paris, <https://doi.org/10.1787/5js4k1fjqkwh-en>. [28]
- OECD (2014), *Regulatory Enforcement and Inspections*, OECD Best Practice Principles for Regulatory Policy, OECD Publishing, Paris, <https://doi.org/10.1787/9789264208117-en>. [53]
- OECD (2014), *The Governance of Regulators*, OECD Best Practice Principles for Regulatory Policy, OECD Publishing, Paris, <https://doi.org/10.1787/9789264209015-en>. [41]

- OECD (2012), *Recommendation of the Council on Regulatory Policy and Governance*, OECD Publishing, Paris, <https://doi.org/10.1787/9789264209022-en>. [25]
- OECD (2010), *Post-Public Employment: Good Practices for Preventing Conflict of Interest*, OECD Publishing, Paris, <https://doi.org/10.1787/9789264056701-en>. [139]
- OECD (2010), *Risk and Regulatory Policy: Improving the Governance of Risk*, OECD Reviews of Regulatory Reform, OECD Publishing, Paris, <https://doi.org/10.1787/9789264082939-en>. [22]
- OECD (2003), *Recommendation of the Council on OECD Guidelines for Managing Conflict of Interest in the Public Service*, <http://OECD/LEGAL/0316>. [45]
- OECD (2002), *Regulatory Policies in OECD Countries: From Interventionism to Regulatory Governance*, OECD Reviews of Regulatory Reform, OECD Publishing, Paris, <https://doi.org/10.1787/9789264177437-en>. [12]
- OECD/KDI (2021), *Case Studies on the Regulatory Challenges Raised by Innovation and the Regulatory Responses*, OECD Publishing, Paris, <https://doi.org/10.1787/8fa190b5-en>. [10]
- Parker, C. (2002), *The Open Corporation: Effective Self-regulation and Democracy*, Cambridge University Press, Cambridge, <https://doi.org/10.1017/cbo9780511550034.001>. [135]
- Patanakul, P. and J. Pinto (2014), “Examining the roles of government policy on innovation”, *The Journal of High Technology Management Research*, Vol. 25/2, pp. 97-107, <https://doi.org/10.1016/j.hitech.2014.07.003>. [17]
- PDPA (2013), “Advisory Guidelines on Key Concepts in the Personal Data Protection Act”, <https://www.pdpc.gov.sg/guidelines-and-consultation/2020/03/advisory-guidelines-on-key-concepts-in-the-personal-data-protection-act>. [99]
- Perarnaud, C. et al. (2022), ‘*Splinternets*’: *Addressing the renewed debate on internet fragmentation*, Parlement Européen; Panel for the Future of Science and Technology. (STOA), <https://shs.hal.science/halshs-03721685v1/document> (accessed on 28 February 2025). [66]
- Pollman, E. (2019), “Tech, Regulatory Arbitrage, and Limits”, *European Business Organization Law Review*, Vol. 20/3, <https://doi.org/10.1007/s40804-019-00155-x>. [87]
- Rees, J. (1997), “Development of Communitarian Regulation in the Chemical Industry”, *Law & Policy*, Vol. 19/4, pp. 477-528, <https://doi.org/10.1111/1467-9930.00036>. [37]
- Ren, S., H. Sun and T. Zhang (2021), “Do environmental subsidies spur environmental innovation? Empirical evidence from Chinese listed firms”, *Technological Forecasting and Social Change*, Vol. 173, p. 121123, <https://doi.org/10.1016/j.techfore.2021.121123>. [107]
- Sarliève, P. et al. (2025), “A mapping tool for digital regulatory frameworks: Including a pilot on efforts to regulate AI”, *OECD Regulatory Policy Working Papers*, No. 23, OECD Publishing, Paris, <https://doi.org/10.1787/1cdad902-en>. [11]
- Selin, J. (2016), “From self-regulation to regulation - An analysis of gambling policy reform in Finland”, *Addiction Research and Theory*, Vol. 24/3, <https://doi.org/10.3109/16066359.2015.1102894>. [152]

- Senden, L. et al. (2015), *Mapping Self- and Co-regulation Approaches in the EU Context: Explorative Study for the European Commission, DG Connect*, European Commission, <https://dspace.library.uu.nl/handle/1874/327305> (accessed on 14 January 2025). [73]
- Short, J. and M. Toffel (2010), "Making self-regulation more than merely symbolic: The critical role of the legal environment", *Administrative Science Quarterly*, Vol. 55/3, <https://doi.org/10.2189/asqu.2010.55.3.361>. [33]
- Sinclair, D. (1997), "Self-regulation versus command and control? Beyond false dichotomies", *Law and Policy*, Vol. 19/4, <https://doi.org/10.1111/1467-9930.00037>. [3]
- Singapore (2012), "Personal Data Protection Act 2012", <https://sso.agc.gov.sg/Act/PDPA2012?WholeDoc=1#Sc1->. [97]
- Stavins, R. (2000), "Market-Based Environmental Policies", in *Public Policies for Environmental Protection*, Routledge, London. [100]
- Stavis, N. (1990), "Incentive-based environmental regulation: A new era from an old idea?", *Ecology Law Quarterly*, Vol. 18, https://www.researchgate.net/publication/5065794_Incentive-based_environmental_regulation_A_new_era_from_an_old_idea (accessed on 12 February 2025). [104]
- Stiglitz, Joseph. E; Rosengard, J. (2015), *Economics of the public sector, Fourth international student edition*. [101]
- Sun, Z. et al. (2021), "The impact of heterogeneous environmental regulation on innovation of high-tech enterprises in China: mediating and interaction effect", *Environmental Science and Pollution Research*, Vol. 28/7, pp. 8323-8336, <https://doi.org/10.1007/s11356-020-11225-w>. [70]
- Tami, S., C. Druehl and R. DeNault (2025), *SEC Announces Possible Last Wave of Off-Channel Communications Enforcement Actions*, <https://www.whitecase.com/insight-alert/sec-announces-possible-last-wave-channel-communications-enforcement-actions> (accessed on 20 May 2025). [138]
- Think Digital Partners (2025), *A Decade After Breaking Up Big IT, UK Government Confronts New Digital Dependencies*, <https://www.thinkdigitalpartners.com/news/2025/12/01/a-decade-after-breaking-up-big-it-uk-government-confronts-new-digital-dependencies>. [114]
- U.S. (1934), "Security Exchange Act", <https://www.govinfo.gov/content/pkg/COMPS-1885/pdf/COMPS-1885.pdf>. [51]
- UK (n.d.), *G-Cloud 11 goes live with 4,200 suppliers*, <https://www.crowncommercial.gov.uk/news/g-cloud-11-goes-live-with-4200-suppliers>. [113]
- UK (n.d.), *G-Cloud 15*, <https://www.crowncommercial.gov.uk/agreements/RM1557.15>. [112]
- US (n.d.), *America's Seed Fund*, <https://www.sbir.gov/>. [111]
- Verbruggen, P. (2013), "Gorillas in the closet? Public and private actors in the enforcement of transnational private regulation", *Regulation & Governance*, Vol. 7/4, pp. 512-532, <https://doi.org/10.1111/rego.12026>. [1]
- Verbruggen, P. (2009), "Does Co-Regulation Strengthen EU Legitimacy?", *European Law Journal*, Vol. 15/4, pp. 425-441, <https://doi.org/10.1111/j.1468-0386.2009.00471.x>. [118]

- Wahyuningtyas, S. (2019), “Self-regulation of online platform and competition policy challenges: A case study on Go-Jek”, *Competition and Regulation in Network Industries*, Vol. 20/1, <https://doi.org/10.1177/1783591719834864>. [63]
- Walters, D. and H. Wiseman (2022), “Self-Regulation in the Cradle: The Role of Standards in Emerging Industries”, *SSRN Electronic Journal*, <https://doi.org/10.2139/ssrn.4226081>. [40]
- World Economic Forum (2020), *Agile Regulation for the Fourth Industrial Revolution: A Toolkit for Regulators*, https://www3.weforum.org/docs/WEF_Agile_Regulation_for_the_Fourth_Industrial_Revolution_2020.pdf (accessed on 10 February 2025). [89]
- Yi, M. et al. (2019), “The heterogeneous effects of different environmental policy instruments on green technology innovation”, *International Journal of Environmental Research and Public Health*, Vol. 16/23, <https://doi.org/10.3390/ijerph16234660>. [69]